

Path-Constrained Haptic Motion Guidance via Adaptive Phase-Based Admittance Control

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Abstract—Robots have surpassed humans in terms of strength and precision, yet humans retain an unparalleled ability for decision-making in the face of unpredictable disturbances. This article aims to combine the strengths of both entities within a singular task: human motion guidance under strict geometric constraints, particularly adhering to predetermined paths. To tackle this challenge, a modular haptic guidance law is proposed that takes the human-applied wrench as an input. Using an auxiliary variable called phase, the generated desired motion is guaranteed to consistently adhere to the constraint path. It is demonstrated how the guidance policy can be generalized into physically interpretable terms, adjustable either prior to initiating the task or dynamically while the task is in progress. Additionally, an illustrative guidance adaptation policy is showcased that takes into account the human's manipulability. Leveraging passivity analysis, potential sources of instability are pinpointed, and subsequently, overall system stability is ensured by incorporating an augmented virtual energy tank. Lastly, a comprehensive set of experiments, including a 20-participant user study, explores various aspects of the approach

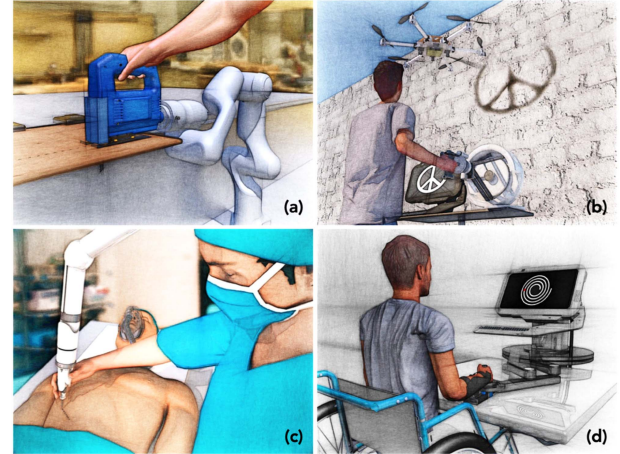


Fig. 1. Illustrative instances of path-constrained scenarios wherein the human guidance on motion profile is pivotal.

Received 3 June 2024; revised 5 September 2024; accepted 4 November 2024. Date of publication 24 December 2024; date of current version 21 January 2025. This work was supported in part by the EU Horizon 2020 ReconCycle project under Grant 871352, in part by the BMBF Project 6G-life under Grant 16KISK002, and in part by the Lighthouse Initiative Geriatrics Project X under Grant 5140951. The work of Petr Svarny was supported by the OP VVV MEYS-funded project under Grant CZ.02.1.01/0.0/0.0/16_019/0000765. The work of Matej Hoffmann was supported by the Czech Science Foundation (GA CR) under Grant 20-24186X. This article was recommended for publication by Associate Editor C. D. Santina and Editor P. R. Giordano upon evaluation of the reviewers' comments. (*Corresponding author: Erfan Shahriari.*)

This work involved human subjects in its research. Approval for all ethical and experimental procedures and protocols was granted by the Ethics Commission of the Technical University of Munich under Application No. 744/21 S-SR.

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This article has supplementary downloadable material available at <https://doi.org/10.1109/TRO.2024.3521861>, provided by the authors.

Digital Object Identifier 10.1109/TRO.2024.3521861

in practice, encompassing both technical and usability considerations.

Index Terms—Adaptive control, motion planning, passivity-based control, physical human-robot interaction.

I. INTRODUCTION

ROBOTS have improved our lives by handling tasks that are burdensome for humans. They can exhibit precision surpassing human capabilities, such as executing intricate motions, or possess substantial strength, allowing them to handle potentially heavy tools. However, there are tasks where human presence is crucial. For instance, in situations where the task is nondeterministic or susceptible to disturbances, human decision-making becomes indispensable. This article focuses on a task requiring expertise from both parties: the robot's precision and the human's guidance.

Consider a scenario where an object (e.g., a tool) must traverse a constraint path. Furthermore, suppose that the motion characteristics should be determined in accordance with a task objective. In this instance, the robot's precision could be utilized to maintain the object on the constraint path, while the human's observations and decision-making prowess could be employed to ascertain the motion attributes along the path. Fig. 1 depicts several illustrations. The object might be a jigsaw machine intended for cutting a wooden block exclusively along a predetermined route. As shown in Fig. 1 (case a), a robot

can secure the machine on the predetermined path, while a human can evaluate the cutting quality and guide the machine's movement along the path. In a comparable scenario (case b), the object could be the handle of a haptic interface employed for remotely controlling a drone executing graffiti on a wall. While the interface's motion can be confined to the intended graffiti shape, the human can assess the drawing quality and subsequently determine the motion along the shape. Fig. 1 (case c) presents an additional scenario where a surgical incision needs to be executed along a designated line on a patient's body, while any deviation from the line might endanger other organs. In this situation, a robot can provide the necessary precision, while the surgeon can establish the speed at which the robot should move the surgical scalpel. Lastly, Fig. 1 (case d) shows a rehabilitation scenario in which, for optimal therapy results, it is vital that the patient's limb adheres to a specific path, aiming to activate particular muscles. While a rehabilitation robot can impose the motion to adhere to the therapeutic path strictly, the patient can actively guide the motion along the path based on their individual abilities. In all the examples mentioned, a robot provides the necessary precision to remain on a restricted path while the human guides the motion by physically interacting with the robot. As a result, the addressed problem is as follows.

Problem Statement: In tasks requiring movement along constraint paths, the objective is to enable the human to guide the motion haptically, i.e., by applying wrenches, while the robot ensures adherence to the path's constraints.

A. Related Works

Considering the stated problem, the related work in the literature can be divided into three main groups, which in turn motivate the three main theoretical contributions of the proposed approach.

1) *Path-Constrained Haptic Motion Guidance:* Numerous works have focused on developing robot guidance policies based on the human's applied wrench. The interaction wrench can be used to convey the guidance intention for choosing among predefined motion policies [1], [2], to adjust predefined trajectories [3], [4], or to generate the robot's desired motion. For the latter, the most well-established approach is through admittance controllers [5]. A comprehensive overview of the application of admittance control in scenarios involving physical human–robot interaction can be found in [6]. More recent works in admittance control for motion guidance include modifying robot compliance near singularities [7], reducing peak contact force during collisions [8], managing cluttered environments [9], incorporating collision detection [10], and handling uncertainties like unknown payloads [11]. While admittance control forms the basis for the guidance policy proposed in this article, a fundamental aspect to consider is the incorporation of geometric constraints.

Numerous works in the literature have focused on constraining the robot motion. A well-known approach is the use of virtual fixtures, as introduced in [12], where a repulsive wrench normal to a predefined surface is virtually applied to the robot. Yet, as the stated objective is to keep the robot on a path rather than

repel it, the adapted concept of virtual fixtures—known as virtual guides—becomes more relevant. One of the earliest works in this context [13], defined desired motion directions based on a predefined path and adjusts admittance control gains accordingly. However, this method has drawbacks. First, altering gains based on path direction does not guarantee that the output trajectory remains on the constraint path, especially with frequent changes or slight communication delays. Second, robot deviations from the path due to imperfect motion control pose challenges in determining the suitable point on the path for redirection. This challenge is shared by approaches like [14] and [15], where continuous assessment of the robot pose in relation to the path is necessary for adjusting motion control gains.

The concept of virtual guides is expanded into virtual mechanisms, initially introduced in the context of teleoperation in [16]. This involves treating the motion setpoint as a mass-less particle positioned on the constraint path and virtually attached to the robot end-effector through a spring-damper system. When the human applies the wrench to the robot, the particle glides along the path. The virtual mechanism has become a well-established approach for path-constrained motion guidance scenarios [17], [18], [19] and shares fundamental similarities with the approach presented in this article. In both approaches, the particle's location along the path is determined by an auxiliary scalar variable, known as phase, which spans values between 0 and 1. As a result, the motion setpoint is consistently determined by identifying the phase value. A fundamental difference from the method proposed in this article is that, in the virtual mechanism approach, the phase trajectory is dependent on the robot pose. Hence, the robot dynamics or motion-following capability can influence the guidance behavior. In the proposed approach, however, the phase progress is determined based on the human's applied wrench through a specialized admittance law, allowing for freely shaping the guidance behavior independently of the robot dynamics or motion controller.

Phase¹ has also been used in other well-established path-centric approaches such as dynamic movement primitives (DMPs) [21]. While the original formulation of DMPs is not intended for online guidance of motion, some work has integrated admittance policies with DMPs for this purpose [22], [23], [24]. However, in none of these approaches is the admittance law derived in the phase domain, which would guarantee that the generated desired trajectory adheres to the constraint path. Throughout the article, the proposed phase-based admittance law is compared with dynamic movement primitives and virtual mechanisms, highlighting the advantages and disadvantages of each method.

2) *Manipulability-Based Guidance Adaptation:* The involvement of a human, a nonuniform entity, in the control loop mandates the adaptability of control parameters to suit the specific needs of each individual human operator. This adaptation can occur either before initiating the guidance task or dynamically during its execution. Several of the aforementioned guidance policies possess this adaptation capability, aligned with diverse objectives. For instance, in [5] and [25], the guidance

¹In the literature, phase may be denoted differently; for instance, as in [20].

parameters are adjusted to robustly handle disturbances and model uncertainties. In [7], parameters are modified in critical areas such as configurations near singularities, joint limits, and workspace boundaries. Admittance control parameter adaptation is frequently employed to alter the control policy, transitioning from autonomous to transparent control, as shown in examples like [11], [26], [27], and [28]. In [29], guidance parameter adaptation is based on online fatigue detection during rhythmic arm movement rehabilitation. Additionally, in [3] and [30] admittance control gain adaptation laws are proposed based on the robot's contact situation with the environment. In [31], the admittance controller gains are adapted based on the perceived human steering intention through interaction force.

Most adaptive admittance control policies focus on shaping the damping component of the control law. For instance, to determine the optimal damping leading to minimal jerk, a reinforcement learning algorithm and a neural network are employed in [32] and [33], respectively. Similarly, in [34], a fuzzy interface system, and in [35] a radial basis function network are utilized to update damping parameters. Guidance damping is adjusted based on human muscle activation in [36], estimated human intention in [37], and task-dependent policies in [38]. The admittance type of the proposed guidance law provides the modularity necessary for independently adapting the guidance dynamics. Accordingly, this article first demonstrates how the parameters of the phase-based admittance law can be represented in a physically interpretable manner. Following this, an adaptation policy is proposed based on the estimated configuration-based guiding capability of the human.

The methodology used for evaluating human ergonomics during the guiding task is based on the concept of manipulability, initially introduced for robot arms in [39] and [40] and later extended to the human body in [41], [42]. Manipulability can be visualized as an ellipsoid, representing the effectiveness of force/motion mapping between different spaces (e.g., operational and joint spaces of an arm). A common use-case scenario for manipulability in the literature is human-robot physical interaction. In [43] and [44], the robot task frame was assigned based on the direction of the human manipulability ellipsoid. Similarly, in [45], the robot motion controller gains are adjusted based on the configuration of the manipulability ellipsoid. An assistive robot control scheme that aids the human in applying forces in problematic directions is presented in [46]. This approach is later extended to consider muscular space ergonomics, focusing on human muscular force manipulability in [47] and [48]. The proposed guidance adaptation policy in this article is based on evaluating the human manipulability ellipsoid with respect to the constraint path. In the experimental section, a state-of-the-art vision-based methodology is utilized to demonstrate the practical implementation of this exemplary adaptation law.

3) *Passivity-Based Adaptive Interaction Control*: Integrating an adaptive interaction control policy necessitates a comprehensive analysis of the system's stability. An increasingly popular approach for this purpose is the passivity analysis method, where the system is modeled as a modular energy-based network of subsystems to identify potentially stability-violating

components. The earliest application of passivity to adaptive robot controllers dates back to the 1980s in [49], followed by [50] and [51], which are among the pioneering instances of passivity-based adaptive motion controllers. In [52], passivity analysis is employed to develop a guidance method that enables humans to modify learned trajectories. Furthermore, in [53], a passivity-based admittance controller is proposed to enhance the robot's ability to interact more naturally with an unknown environment.

A common approach to ensuring system passivity involves utilizing a virtual energy tank. Originating in the teleoperation field [54], energy tanks have found widespread application in interaction scenarios. For instance, in [55], a virtual energy tank is combined with a variable impedance controller to maintain overall system passivity. In [56], energy tanks sense the human guidance level while the robot follows passively, and in [57], conventional tank dynamics are extended to adapt to manipulation behavior. Moreover, recent works such as [3], [58], [59], [60], and [61], ensure stability by adjusting admittance controller based on observed passivity-violating energy. Virtual tanks are also employed in this article to ensure stability. In this context, the tank-based adaptive admittance control method, initially introduced in [62] and later applied in [63], [64], and [65], represents the most closely related approach. However, as previously discussed, the objective here is not to utilize conventional admittance control for the guidance policy. Therefore, a novel energy storage function and an energy tank are formulated for the system, addressing the potential passivity-violating effect of the guidance adaptation. Subsequently, a Lyapunov analysis is employed to reverify the overall system stability. Finally, an experimental setup is devised to practically illustrate how the configuration of the tank affects the system's behavior in an exemplary collision scenario.

Contributions: Drawing upon the selected relevant literature highlighted above, this article's contributions can be summarized as follows.

- 1) Development of a haptic motion guidance law that ensures consistent adherence of the generated motion to a constraint path while allowing for the shaping of guidance characteristics independently of the robot dynamics or motion control performance.
- 2) Integration of an illustrative ergonomic guidance adaptation policy that takes into account the manipulability of the human operator.
- 3) Ensuring the overall stability of the system through a thorough passivity analysis and the derivation of a virtual energy tank tailored to the proposed guidance policy.
- 4) Implementation and analysis of comprehensive experiments to explore various facets of the approach, covering technical aspects and usability, including a 20-participant user study.

B. Structure

In the following sections, initially, the theoretical framework is presented, consisting of four key components illustrated in Fig. 2. It begins with the basics, covering the manipulator model,

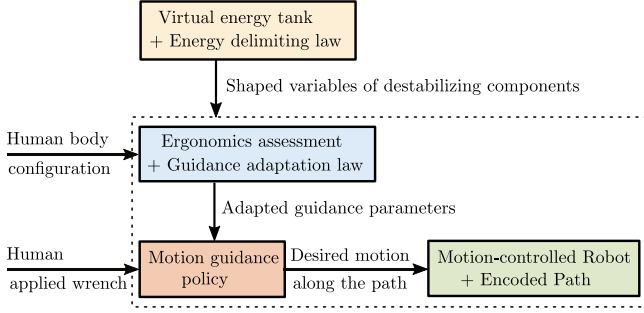


Fig. 2. Approach structure; the concepts associated with modeling and motion control (green), guidance law (red), guidance adaptation (blue), and stability (yellow) are explained in Sections II, III, IV, and V, respectively.

motion control, and the constraint path in Section II. Next, in Section III, the proposed motion guidance policy is introduced, which takes the human-applied wrench as input and generates a corresponding motion command as output. Following that, Section IV presents the guidance adaptation policy based on human manipulability as an ergonomics evaluation metric. Concluding the theoretical aspects, Section V conducts a passivity analysis of the proposed system followed by formulating a virtual energy tank to ensure stability. Additionally, this section revisits the proposed tank-based approach, employing Lyapunov stability analysis. Moving on to Section VI, the article presents the outcomes of five distinct sets of experiments, each focusing on a unique aspect of the proposed approach. Finally, Section VII concludes this article.

II. FUNDAMENTALS: MANIPULATOR MOTION CONTROL

Consider an n -DoF robot manipulator that may carry an object and is intended to move along a predefined path under human guidance (see Fig. 3). Its dynamics in Cartesian space—while holding the object rigidly—are well known to be [66]

$$M_C(q)\ddot{x} + C_C(q, \dot{q})\dot{x} + f_g(q) = f_{in} + f_h \quad (1)$$

where $q, \dot{q} \in \mathbb{R}^n$ are the joint position and velocity, and $\dot{x}, \ddot{x} \in \mathbb{R}^m$ are the end-effector velocity and acceleration in the m -dimensional Cartesian space. The terms $M_C(q), C_C(q, \dot{q}) \in \mathbb{R}^{m \times m}$ denote the Cartesian space representation of the mass matrix, and the Coriolis and centrifugal matrix, respectively, of the robot rigidly holding the object. Finally, $f_g(q), f_{in} \in \mathbb{R}^m$ are the vectors of gravity and the control input, respectively, expressed in the Cartesian space, and $f_h \in \mathbb{R}^m$ is the wrench applied by the human on the robot. Please note that for the sake of completeness, if the robot is holding an object, it is assumed that the mass matrix, the Coriolis, and centrifugal matrix, and the gravity vector all incorporate the object dynamics effect, as well. Furthermore, for the sake of simplicity and without sacrificing generality, the frictional effect is omitted in (1).

A. Motion Controller

The manipulator is supposed to move along the predetermined path while compensating for the object's dynamics. Considering continuous interaction with the human operator, the following

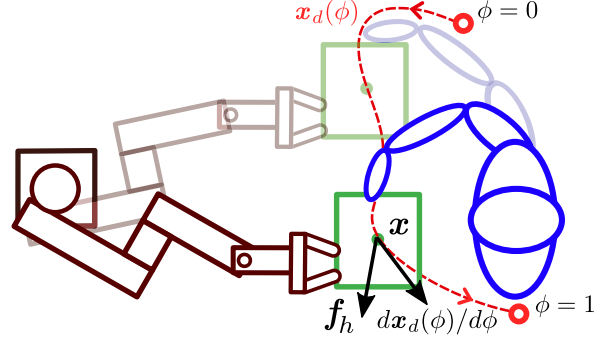


Fig. 3. Human guides a robot (carrying an object) along a predefined path. The applied wrench f_h directs the desired motion through the encoded path $x_d(\phi)$, from $\phi = 0$ to $\phi = 1$.

impedance control law with gravity compensation [67] is used for motion control

$$\begin{aligned} f_{in} &= K_C \tilde{x} + D_C \dot{\tilde{x}} + M_C(q)\ddot{x}_d + C_C(q, \dot{q})\dot{x}_d + f_g(q) \\ \tilde{x} &= x_d - x \end{aligned} \quad (2)$$

where $K_C, D_C \in \mathbb{R}^{m \times m}$ are the desired Cartesian stiffness and damping of the impedance controller and $x_d \in \mathbb{R}^m$ is the desired robot pose. System (1) with controller (2) has the following closed-loop dynamics:

$$M_C(q)\ddot{\tilde{x}} + C_C(q, \dot{q})\dot{\tilde{x}} + D_C \dot{\tilde{x}} + K_C \tilde{x} + f_h = 0. \quad (3)$$

The approach for guiding motion along a path is to empower the human operator to continuously determine which point of the path should be the current setpoint x_d for the motion controller. To achieve this, an auxiliary variable is employed.

B. Encoded Constraint Path

To continuously assign the setpoint along the constraint path for the motion controller (2), one approach involves encoding all path points using an auxiliary variable phase $\phi \in [0, 1]$. By doing so, the path and, accordingly, the setpoint is transformed into a continuous function of the phase variable, i.e., $x_d(\phi)$, where $\phi = 0$ and $\phi = 1$ correspond to the initial and the final points of the path, respectively (see Fig. 3). This function embodies the geometric representation of the constraint path, devoid of temporal information integration.

Encoding a path using an auxiliary scalar variable has been employed in several well-established contexts, such as dynamic movement primitives [21] and virtual guides [68]. The encoding process can be done either by manually deriving the path as a function of phase (e.g., via a linear mapping) or by collecting all the path points and using function-fitting techniques such as Gaussian methods [69], [70], [71]. With a phase-encoded path, the problem of determining the setpoint x_d is simplified to determining the phase value ϕ . In essence, by enabling the human operator to ascertain the phase trajectory $\phi(t)$ and integrating it into the encoded function $x_d(\phi)$, the desired motion trajectory $x_d(t)$ can be attained. The subsequent chapter outlines an approach illustrating how the human operator can determine

the phase trajectory—thus guiding robot motion—by exerting wrench on the robot.

III. HUMAN-GUIDED MOTION GENERATOR

The human operator is expected to consistently determine the setpoint $\mathbf{x}_d$ on the predefined path by applying wrench to the robot. To do so, a policy is derived which continuously generates the phase trajectory $\phi(t)$ according to the human-applied wrench $\mathbf{f}_h$.

A. Force-Driven Phase Evolution Control

The core idea is to empower the human operator to direct the phase progression through the contact wrench $\mathbf{f}_h$. The assumption is that the human's guiding intention correlates with the magnitude of $\mathbf{f}_h$ and its alignment with the predefined path.² This relationship can be articulated by introducing the steering force variable $f_s \in \mathbb{R}$ as the projection of the sensed wrench $\mathbf{f}_h$ onto the tangent vector to the encoded path $\mathbf{x}_d(\phi)$ (i.e., $d\mathbf{x}_d(\phi)/d\phi$) at each point³ as follows:

$$f_s = \mathbf{f}_h^T \frac{d\mathbf{x}_d(\phi)}{d\phi}. \quad (4)$$

The steering force f_s then propels the phase evolution through the following nonlinear admittance law:

$$m_\phi \ddot{\phi} + d_\phi \dot{\phi} + c_\phi \dot{\phi}^2 = f_s \quad (5)$$

with $m_\phi, d_\phi \in \mathbb{R}_{>0}$ representing the phase-based mass and damping coefficients, and $c_\phi \in \mathbb{R}$ denoting the coefficient for the nonlinear term $\dot{\phi}^2$. As shown in the next section, the rationale for the nonlinear form in (5) is to provide an intuitive physical representation, simplifying parameter tuning.

The phase-based admittance law (5) adjusts the phase value ϕ , either by increasing or decreasing it. To constrain ϕ within the range $[0, 1]$ for iterative tasks (e.g., circular path traversal), the phase resets to 0 or 1 upon reaching the boundaries. Conversely, for nonrepetitive tasks (e.g., linear segment motion), one method is to increase d_ϕ as ϕ approaches the borders or, alternatively, to saturate the phase value.

B. Generalized Admittance Parameters

The phase-based admittance (5) serves to transform the applied wrench $\mathbf{f}_h$ into the intended motion along the predefined path. The phase-based parameters m_ϕ, d_ϕ , and c_ϕ govern the characteristics of this transformation, affecting the force required for a specific desired motion profile. Yet, adjusting them might lack intuitiveness, as (5) primarily addresses phase dynamics rather than direct motion aspects. To address this, auxiliary parameters $m_a, d_a \in \mathbb{R}_{>0}$ are introduced. These parameters endow the wrench-to-motion mapping with the familiar properties of a mass-damper system. To achieve this, the phase-based admittance parameters are defined (refer to the derivation

in Appendix A) as

$$m_\phi = m_a \frac{d\mathbf{x}_d^T(\phi)}{d\phi} \frac{d\mathbf{x}_d(\phi)}{d\phi} \quad (6)$$

$$d_\phi = d_a \frac{d\mathbf{x}_d^T(\phi)}{d\phi} \frac{d\mathbf{x}_d(\phi)}{d\phi} \quad (7)$$

$$c_\phi = m_a \frac{d^2\mathbf{x}_d^T(\phi)}{d\phi^2} \frac{d\mathbf{x}_d(\phi)}{d\phi}. \quad (8)$$

As a result, the guidance policy (5) can be effectively tailored by adjusting the intuitive admittance mass and damping coefficients m_a and d_a with units $[\text{kg}]$, $[\text{kg} \cdot \text{m}^2]$ and $[\text{N} \cdot \text{s}/\text{m}]$, $[\text{N} \cdot \text{m} \cdot \text{s}/\text{rad}]$, respectively. When the contact wrench aligns with the path tangent vector, the phase-based guidance law (5) transforms into

$$m_a \ddot{\mathbf{x}}_d + d_a \dot{\mathbf{x}}_d = \mathbf{f}_h. \quad (9)$$

Please note that the nonlinear term $c_\phi \dot{\phi}^2$ in (5) is essential to achieve the intuitive representation (9) using the parameter transformation relations (6)–(8); see Appendix A. Appendix B compares the proposed guidance law with the well-established approach of dynamic movement primitives [21], which shares fundamental similarities.

The phase progress control (4)–(8) enables the human operator to determine the desired motion along the constraint path. By adjusting the admittance mass and damping coefficients m_a and d_a , the responsiveness of this guidance law can be fine-tuned. Lower values of these coefficients result in faster motions for a given wrench, while higher values lead to slower motion. Depending on task requirements, admittance coefficients can be preset to suitable values or alternatively dynamically adapt according to the task situation. The upcoming section presents an exemplar adaptation policy that takes into account the estimation of the human operator's configuration-based guiding capability.

IV. MOTION GUIDANCE ADAPTATION

The adaptation of guidance law parameters can be approached through various policies. The proposed approach here is built on the rationale that continuous interaction between the human operator and the robot for guidance can result in different human body configurations and varying abilities to guide effectively. For instance, certain configurations might make applying the guidance wrench more challenging, while in other configurations, moving along with the robot could become difficult. The suggested approach to parameter adaptation stems from this insight, leveraging the concept of manipulability to assess the human's guiding capability and adjust the admittance mass accordingly.

A. Human Guiding Capability Along a Path

One method to evaluate guiding capability is by estimating the required joint torques and motions for both the guiding wrench $\mathbf{f}_h$ and the resulting guided trajectory $\mathbf{x}_d(t)$. A well-established approach for this purpose is the manipulability concept [40]. Given $\mathbf{J}_h(\mathbf{q}_h) \in \mathbb{R}^{m \times n_h}$ as the Jacobian matrix of an n_h -DoF human body, a manipulability ellipsoid can be

²For this assumption, the human operator needs to know or be instructed about the current path direction and how to apply the interaction wrench.

³Please note that, the translational and rotational components could have been treated separately, for example, by using different individual weights.

constructed using a symmetric positive definite core matrix $\Lambda \in \mathbb{R}^{m \times m}$, defined as

$$\Lambda(\mathbf{q}_h) = \mathbf{J}_h(\mathbf{q}_h) \Upsilon \mathbf{J}_h^T(\mathbf{q}_h) \quad (10)$$

where $\Upsilon \in \mathbb{R}^{n_h \times n_h}$ may assume various forms depending on the chosen approach [72]. This ellipsoid can be visualized in the Cartesian space by $\mathbf{x}_{\text{ell}} \in \mathbb{R}^m$ satisfying

$$(\mathbf{x}_{\text{ell}} - \mathbf{x}_h)^T \Lambda^{-1}(\mathbf{q}_h) (\mathbf{x}_{\text{ell}} - \mathbf{x}_h) = 1 \quad (11)$$

with $\mathbf{x}_h \in \mathbb{R}^m$ being the human interaction pose located at the ellipsoid's center. The ellipsoid's principal radii are along the eigenvectors $\boldsymbol{\nu}_i \in \mathbb{R}^m$ of the core matrix $\Lambda(\mathbf{q}_h)$. Their lengths $r_i \in \mathbb{R}_{\geq 0}$ can be determined as

$$r_i = \sqrt{\lambda_i} \quad (12)$$

with $\lambda_i \in \mathbb{R}_{\geq 0}$ being the eigenvalues of $\Lambda(\mathbf{q}_h)$.

The ellipsoid's radii lengths represent the viability of end-point motions in the Cartesian space along their respective directions. A longer radius length r_i implies smaller joint motions are required to achieve the Cartesian motion along that radius. Orthogonal to this ellipsoid is the force manipulability ellipsoid, whose radii determine the feasibility of applying wrenches along their corresponding directions. When the path follows the longest radius of the force manipulability ellipsoid (i.e., the shortest radius of the manipulability ellipsoid (11)), smaller joint torque values are necessary to produce the same applied wrench magnitudes. Thus, to evaluate the situation at each phase ϕ , the path direction $d\mathbf{x}_d(\phi)/d\phi$ can be compared with the manipulability ellipsoid's principal radii direction and length. To achieve this, the principal radii can be first projected onto the path direction $\boldsymbol{\nu}(\phi) \in \mathbb{R}^m$, resulting in the projected length $l_i \in \mathbb{R}$ according to

$$l_i = |r_i \boldsymbol{\nu}_i^T \boldsymbol{\nu}(\phi)| \quad (13)$$

where

$$\boldsymbol{\nu}(\phi) = \frac{d\mathbf{x}_d(\phi)/d\phi}{\|d\mathbf{x}_d(\phi)/d\phi\|}. \quad (14)$$

Considering $\mathbf{l}_\Lambda = [l_1, l_2, \dots, l_{n_\Lambda}]$ being the vector containing the projection lengths, its norm l is

$$l = \sqrt{\sum_{i=1}^{n_\Lambda} l_i^2}. \quad (15)$$

The projection norm⁴ l always falls between the largest and smallest radii lengths [72]. A higher l value suggests a higher likelihood that the desired path direction aligns with the longest radii of the manipulability ellipsoid. Therefore, the ellipsoid's radii projection norm l can be employed to influence the admittance parameters.

B. Configuration-Based Admittance Shaping

As explained earlier, in each body configuration, the projection norm l can serve as a metric for evaluating the human

operator's guidance capability along various path directions. To incorporate it into the adaptation of admittance parameters, first, a range $[r_{\min}, r_{\max}]$ needs to be defined. Here, $r_{\min}, r_{\max} \in \mathbb{R}_{\geq 0}$ denote the assumed smallest and largest potential values of l , respectively, associated with $m_{a,\max}, m_{a,\min} \in \mathbb{R}_{\geq 0}$, representing the lowest and highest m_a values. For instance, given that the shortest radius of the manipulability ellipsoid becomes 0 during singularities, one possible choice for presetting $r_{\min}$ could be 0. Similarly, considering $\lambda_{\max}$ as the largest possible eigenvalue of the matrix $\Lambda(\mathbf{q}_h)$, $r_{\max}$ may be set to

$$r_{\max} = \sqrt{\lambda_{\max}}. \quad (16)$$

The guidance law can be adjusted in a manner such that when l approaches $r_{\max}$, less interaction wrench from the human is needed to guide the motion. Conversely, when l is nearer to $r_{\min}$, lower velocities result from the interaction wrench. One approach to achieve this is by establishing a linear relationship between the projection norm l and the admittance parameters. For instance, the admittance mass m_a can be tailored through

$$m_a = \begin{cases} m_{a,\max} & l \leq r_{\min} \\ m_{a,\min} + (r_{\max} - l) \frac{\Delta m_a}{\Delta r} & r_{\min} < l < r_{\max} \\ m_{a,\min} & r_{\max} \leq l \end{cases} \quad (17)$$

with $\Delta r = r_{\max} - r_{\min}$ and $\Delta m_a = m_{a,\max} - m_{a,\min}$. Hence⁵, when l is large—suggesting weaker force capacity— m_a (and m_ϕ) is lower, reducing the demand for steering force in (5). Conversely, when l is small—indicating limited motion feasibility along the path— m_a is set higher, resulting in slower desired motions. Please note that a similar approach could be employed to shape the admittance damping parameter d_a .

As shown in the following section, modifying the admittance mass—regardless of the chosen adaptation policy—might lead to potential instability. Hence, apart from assessing the overall system's stability, an approach is presented to ensure stability even during guidance adaptation.

V. SYSTEM STABILIZATION

So far, the overall system consists of a robot with its motion controller and an adaptive phase-based motion generator whose input is produced by a human operator continuously interacting with the robot. This interlinked architecture, with adaptive parameters, necessitates a stability analysis to guarantee that the system remains stable and adheres to the intended behavior. To explore stability, the approach of passivity analysis [49] is employed, which essentially examines the energy flow throughout the entire system by associating energy models with each component and interconnecting them power-conservatively. Subsequently, parts with potentially unbounded, i.e., passivity-violating, energy are identified. Among several approaches, the virtual energy tank augmentation method is utilized to confine passivity-violating energy and ensure the stability of the entire system.

⁴Also referred to as the radius of the manipulability pseudoellipsoid in [72].

⁵it is assumed that $\Delta r \neq 0$, unless $m_{a,\min} = m_{a,\max}$.

A. Passivity Analysis

To examine the passivity, the system is first divided into several subsystems with their associated energy storage functions $S_i \in \mathbb{R}_{\geq 0}$. The summation of the subsystems' storage functions forms the overall storage function $S = \sum S_i$. The system is autonomously passive if S is upper-bounded, which can be ensured by proving [57]

$$\forall t \quad \dot{S}(t) \leq 0. \quad (18)$$

Consider the following energy storage function for the impedance-controlled manipulator (1)–(2) in the presence of the phase-based admittance law (5):

$$S_{\text{sys}} = \frac{1}{2} \dot{\mathbf{x}}^T \mathbf{M}_C(\mathbf{q}) \dot{\mathbf{x}} + \frac{1}{2} \tilde{\mathbf{x}}^T \mathbf{K}_C \tilde{\mathbf{x}} + \frac{1}{2} m_\phi \dot{\phi}^2 \geq 0. \quad (19)$$

The time evolution of (19) becomes

$$\begin{aligned} \dot{S}_{\text{sys}} = & \dot{\mathbf{x}}^T \mathbf{M}_C(\mathbf{q}) \ddot{\mathbf{x}} + \frac{1}{2} \tilde{\mathbf{x}}^T \dot{\mathbf{M}}_C(\mathbf{q}) \dot{\mathbf{x}} + \dot{\mathbf{x}}^T \mathbf{K}_C \tilde{\mathbf{x}} \\ & + m_\phi \ddot{\phi} \dot{\phi} + \frac{1}{2} \dot{m}_\phi \dot{\phi}^2. \end{aligned} \quad (20)$$

Considering the closed-loop dynamics (3) and the phase-based guidance law (5), the time evolution (20) becomes

$$\begin{aligned} \dot{S}_{\text{sys}} = & \dot{\mathbf{x}}^T \underbrace{(-\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{x}} - \mathbf{D}_C \dot{\mathbf{x}} - \mathbf{K}_C \tilde{\mathbf{x}} - \mathbf{f}_h)}_{\mathbf{M}_C(\mathbf{q}) \ddot{\mathbf{x}}} \\ & + \frac{1}{2} \dot{\mathbf{x}}^T \dot{\mathbf{M}}_C(\mathbf{q}) \dot{\mathbf{x}} + \dot{\mathbf{x}}^T \mathbf{K}_C \tilde{\mathbf{x}} \\ & + \underbrace{\left(f_s - d_\phi \dot{\phi} - c_\phi \dot{\phi}^2 \right)}_{m_\phi \ddot{\phi}} \dot{\phi} + \frac{1}{2} \dot{m}_\phi \dot{\phi}^2 \end{aligned} \quad (21)$$

$$\begin{aligned} = & \frac{1}{2} \dot{\mathbf{x}}^T \left(\dot{\mathbf{M}}_C(\mathbf{q}) - 2\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}) \right) \dot{\mathbf{x}} - \dot{\mathbf{x}}^T \mathbf{D}_C \dot{\mathbf{x}} \\ & - \dot{\mathbf{x}}^T \mathbf{f}_h + f_s \dot{\phi} - d_\phi \dot{\phi}^2 - c_\phi \dot{\phi}^3 + \frac{1}{2} \dot{m}_\phi \dot{\phi}^2 \end{aligned} \quad (22)$$

$$\begin{aligned} = & -\dot{\mathbf{x}}^T \mathbf{D}_C \dot{\mathbf{x}} + \dot{\mathbf{x}}^T \mathbf{f}_h - \dot{\mathbf{x}}_d^T \mathbf{f}_h + f_s \dot{\phi} \\ & - d_\phi \dot{\phi}^2 - c_\phi \dot{\phi}^3 + \frac{1}{2} \dot{m}_\phi \dot{\phi}^2 \end{aligned} \quad (23)$$

where (23) is achieved by taking the skew-symmetric property of the matrix $(\dot{\mathbf{M}}_C(\mathbf{q}) - 2\mathbf{C}_C(\mathbf{q}, \dot{\mathbf{q}}))$ in (22) into account. Considering the chain rule

$$\dot{\mathbf{x}}_d = \frac{d\mathbf{x}_d(\phi)}{d\phi} \dot{\phi} \quad (24)$$

and the following equality (see Appendix C for the proof):

$$\dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d = \dot{m}_\phi \dot{\phi}^2 - 2c_\phi \dot{\phi}^3 \quad (25)$$

the storage function's time-derivative in (23) reduces to

$$\begin{aligned} \dot{S}_{\text{sys}} = & -\dot{\mathbf{x}}^T \mathbf{D}_C \dot{\mathbf{x}} + \dot{\mathbf{x}}^T \mathbf{f}_h - \underbrace{\dot{\phi} \frac{d\mathbf{x}_d^T(\phi)}{d\phi} \mathbf{f}_h}_{f_s} + f_s \dot{\phi} \\ & - d_\phi \dot{\phi}^2 - \frac{1}{2} \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d \end{aligned} \quad (26)$$

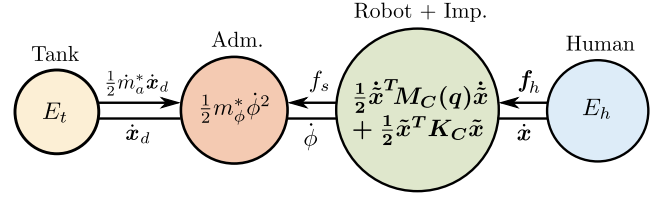


Fig. 4. Power-flow representation of the overall system. The phase-based admittance mass m_ϕ^* is associated with the modified admittance mass m_a^* after tank augmentation.

$$= -\dot{\mathbf{x}}^T \mathbf{D}_C \dot{\mathbf{x}} + \dot{\mathbf{x}}^T \mathbf{f}_h - d_\phi \dot{\phi}^2 + \frac{1}{2} \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d. \quad (27)$$

Finally, considering the positive-definiteness of the damping matrix $\mathbf{D}_C$ and the positivity⁶ of the phase-based admittance damping d_ϕ , the following holds:

$$\dot{S}_{\text{sys}} \leq \dot{\mathbf{x}}^T \mathbf{f}_h + \frac{1}{2} \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d. \quad (28)$$

Considering passivity condition (18), inequality (28) does not prove the system passivity as the powers associated with the admittance mass adaptation $\frac{1}{2} \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d$ and the human physical interaction $\dot{\mathbf{x}}^T \mathbf{f}_h$ are not necessarily always negative; see Fig. 4. Assuming a limited amount of energy E_h corresponding to the human operator, it can be deduced that

$$\dot{E}_h \leq -\dot{\mathbf{x}}^T \mathbf{f}_h \quad (29)$$

$$E_h \geq 0. \quad (30)$$

This assumption, a common premise in the literature, implies that the human would not input an unrestricted amount of energy into the controlled robot system [73]. Note that such an assumption also holds true for any passive environment applying the guiding wrench to the robot. An illustrative situation is presented in Experiments 1 and 3; see Section VI. Yet, this assertion cannot be applied to the power associated with the adaptation of the guidance law. As previously noted, adjusting the admittance mass could potentially lead to passivity violations, and the energy related to the corresponding power $\frac{1}{2} \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d$ must be restricted to ensure autonomous passivity [57]. One way to achieve this is by augmenting a virtual energy tank in the system, as demonstrated next.

B. Augmentation of Virtual Energy Tank

As previously discussed, the change in the admittance mass $\dot{m}_a$ can potentially disrupt passivity. To address this, a virtual energy tank can be augmented to impose an upper bound $\mu \in \mathbb{R}_{>0}$ on the rate of mass variation and consequently constrain the associated passivity-violating power. As a result, the admittance mass m_a is adjusted (refer to Appendix D) to become m_a^* , so that

$$|\dot{m}_a^*| \leq \mu. \quad (31)$$

The phase-based admittance parameters in (5) can then be achieved by employing m_a^* in (6) and (8).

⁶Please note that under this condition, any adaptation of the parameters d_a or d_ϕ would not violate the passivity.

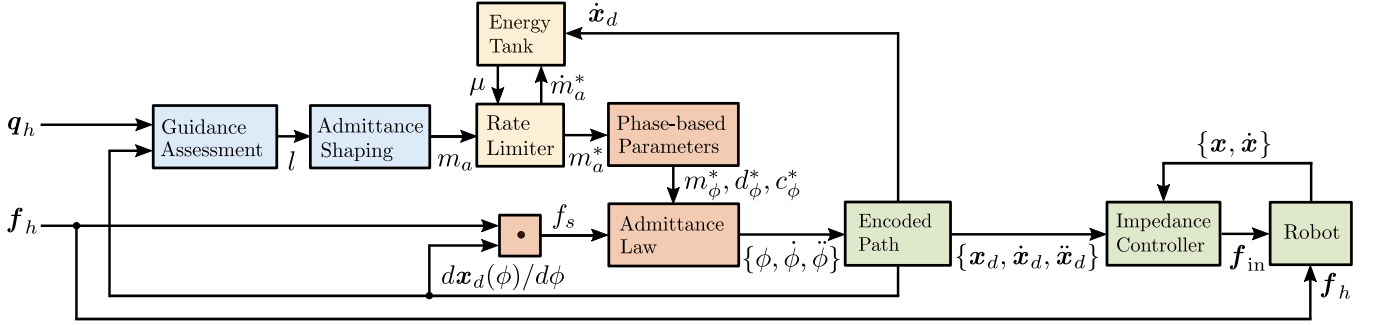


Fig. 5. Proposed control architecture in detail; the blocks associated with the guidance adaptation approach (blue) are explained in Section IV, the stability-ensuring blocks (yellow) are presented in Section V, the motion guidance section (orange) is explained in Section III, and finally, the motion-controlled robot (green) is described in Section II. The compact architecture is depicted in Fig. 2.

The tank energy $E_t \in \mathbb{R}_{>0}$ with initial value $E_{t,0}$ has the following dynamics:

$$\dot{E}_t = -\frac{1}{2} \dot{m}_a^* \dot{x}_d^T \dot{x}_d. \quad (32)$$

Considering (32) for a nonstationary setpoint (i.e., $\dot{x}_d \neq 0$), changes in the admittance mass alter the tank's energy level. An increase in m_a , i.e., a potentially passivity-violating action, depletes the tank energy. Conversely, when m_a decreases, associated with a potentially dissipative effect, the tank energy level increases⁷. By limiting the changes in the admittance mass, these alterations in the tank energy become restricted. In the extreme case, when the tank is about to become empty (i.e., when $E_t \approx 0$), the tank output μ decreases to zero to prevent any further changes in the admittance mass. Accordingly, the following rule can be assigned

$$\mu = \begin{cases} (m_{a,\max} - m_{a,\min}) & \text{if } E_\Delta \leq E_t \\ \frac{1}{2}(m_{a,\max} - m_{a,\min})(1 - \cos(\pi \frac{E_t}{E_\Delta})) & \text{if } 0 \leq E_t < E_\Delta \\ 0 & \text{else} \end{cases} \quad (33)$$

where $E_\Delta \in \mathbb{R}_{>0}$ is the lower energy threshold from which the admittance variation rate gets regulated in case of energy depletion. According to (33), as soon as the tank is empty, μ smoothly⁸ decreases to 0, and considering (31) and (32), the tank does lose any additional energy. Thus, its energy level remains always nonnegative, i.e.,

$$E_t \geq 0. \quad (34)$$

Considering (19), (30), and (34), an overall storage function S_{all} can be defined as

$$S_{\text{all}} = S_{\text{sys}} + E_h + E_t \geq 0. \quad (35)$$

Now, the overall system passivity can be proven according to (28), (29), and (32)

$$\begin{aligned} \dot{S}_{\text{all}} &= \underbrace{\dot{S}_{\text{sys}}}_{\leq \dot{x}^T f_h + \frac{1}{2} \dot{m}_a^* \dot{x}_d^T \dot{x}_d} + \underbrace{\dot{E}_h}_{\leq -\dot{x}^T f_h} + \underbrace{\dot{E}_t}_{=-\frac{1}{2} \dot{m}_a^* \dot{x}_d^T \dot{x}_d} \leq 0. \end{aligned} \quad (36)$$

The power flow map of the system is depicted in Fig. 4. Moreover, the architecture of the entire system is illustrated in Fig. 5.

C. Stability Analysis Following Tank Augmentation

The overall system is demonstrated to be passive. Yet, to examine its Lyapunov stability, the system's state variable is defined as $z = [\tilde{x}^T, \dot{\tilde{x}}^T, \dot{\phi}, e_h, e_t]^T$, where $e_h, e_t \in \mathbb{R}$ denote the states of the human energy and the tank, respectively, with which the energy values E_h and E_t are determined. Moreover, consider e_h^* and e_t^* associated with the energy values $E_h^* = E_h(e_h^*)$ and $E_t^* = E_t(e_t^*)$, such that

$$\forall z \in \mathcal{X} \setminus \{z^*\}, S_{\text{all}}(z) > E_h^* + E_t^* \quad (37)$$

where $\mathcal{X}$ is the space of all possible values for z during the task, and $z^* = [0^T, 0^T, 0, e_h^*, e_t^*]^T$. The candidate Lyapunov function would be

$$V(z) = S_{\text{all}}(z) - E_h^* - E_t^*. \quad (38)$$

Given (19), (37), and (38), it is clear that $V(z^*) = 0$, and

$$\forall z \setminus z^*, V(z) > 0. \quad (39)$$

Thus, considering the continuous differentiability of (38), $V(z)$ is a valid Lyapunov function. Moreover, given (36), it can be observed that $\dot{V}(z) \leq 0$. Therefore, the overall system is Lyapunov stable at $[\tilde{x}^T, \dot{\tilde{x}}^T, \dot{\phi}, e_h, e_t]^T = [0^T, 0^T, 0, e_h^*, e_t^*]^T$. Note that when taking into account the dissipation effect in any of the involved subsystems, (36) transforms into $\dot{S}_{\text{all}} < 0$, resulting in $\dot{V}(z) < 0$. Hence, the overall system becomes asymptotically stable.

To examine the practical effects of augmenting the tank, the system's behavior at equilibrium is examined first. At z^* , $\tilde{x} = 0$ and $\dot{\tilde{x}} = 0$. This implies that the robot's pose and velocity are at the desired values. Furthermore, $\dot{\phi} = 0$ can be interpreted as the operator having guided the robot's desired motion through the applied wrench until the setpoint reaches the intended value. Consequently, there is no longer a necessity for any modification to the phase value. The energy value S_{sys} in (19) can serve as a representation of the extent to which the system state z

⁷An upper limit can be set to prevent overaccumulation of energy [57].

⁸A nonsmooth linear function similar to (17) could have also been used.

deviates from the desired state z^* . The passivity of the system (i.e., $\dot{S}_{\text{sys}} \leq 0$) would have ensured that such a deviation would never be as significant as the initial deviation. Being strictly passive (i.e., $\dot{S}_{\text{sys}} < 0$) would have demonstrated that over time, z would inevitably approach z^* . However, (28) revealed that none of these conditions are necessarily true.

The proposed solution is founded on the principle of ensuring that the potentially positive powers $\dot{x}^T f_h$ and $\frac{1}{2}\dot{m}_a \dot{x}_d^T \dot{x}_d$ in (28) are associated with bounded energy levels. Accordingly, the energy S_{sys} is permitted to transiently increase (indicating a deviation from equilibrium) only up to a certain threshold level defined by the initial values of E_h and E_t . By allowing such deviations, facilitation of the achievement of other task objectives is enabled, namely, the phase evolution due to the applied wrench and the adaptation of the admittance mass. If E_h or E_t become depleted, these objectives would not be met, ensuring that straying too far from the equilibrium state is avoided. Specifically, if the tank energy is drained, the adaptation of the guidance parameter would cease; Experiment 3 illustrates such a scenario. Therefore, a natural question that arises is: How much energy should be allocated for E_h and E_t ? More precisely, what should be the initial value of the tank energy? While this determination could be based on the specific task scenario or facilitated through energy learning algorithms [74], such an investigation is beyond the scope of the current article. Therefore, it is assumed that an appropriate tank initialization policy is utilized.

VI. EXPERIMENTS

This section encompasses five comprehensive sets of experiments aimed at a thorough assessment of the proposed approach. Experiment 1 commences by verifying the phase-based admittance law from (5) under a constant applied force. It then illustrates the generalization of admittance parameters through (6)–(8). Experiment 2 delves into the investigation of the suggested guidance adaptation policy outlined in Section IV. Subsequently, Experiment 3 practically validates the augmentation of the virtual energy tank, particularly in a collision scenario. The initial three sets of experiments elucidate the functioning of each approach component, while the subsequent two sets focus on verifying the proposed phase-based guidance law. First, it is practically compared with the well-established approach of a virtual mechanism in Experiment 4. Finally, Experiment 5 assesses the usability of the proposed guidance method through a dedicated use-case study.

The experiments are conducted employing a 7-DoF Franka robot. The guiding wrench in Experiment 1 originates from the gravity of an attached unknown load (Fig. 6), while in Experiment 4, it is manually set to a constant value of 5N. In contrast, in Experiments 2, 3, and 5, the guiding force is imparted by the human operator through an applied wrench; see Figs. 8, 13, and 15. The external wrench f_h measured by the robot shapes the desired trajectory by generating the phase trajectory according to (4)–(5). The generated phase value is saturated to always remain between 0 and 1. For the implementation of the Cartesian impedance controller (2), the stiffness K_C takes the form of a

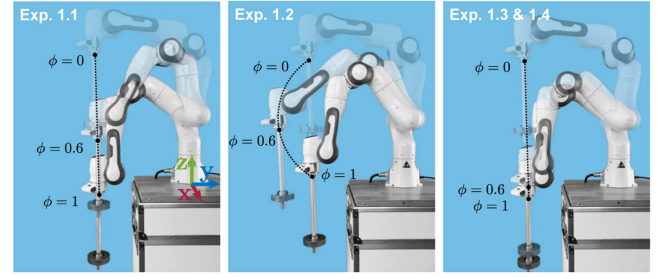


Fig. 6. Experiment 1 scenarios; the attached weight propels the phase ϕ toward 1, causing the setpoint to move along the encoded path. To demonstrate the impact of phase distribution on trajectory, in Exp. 1.1 and Exp. 1.2, the phase is uniformly spread across the path, whereas in Exp. 1.3 and Exp. 1.4, an exponential phase distribution is chosen.

diagonal matrix with elements $\{1000, 1000, 1000, 50, 50, 50\}$ [N/m, Nm/rad], and the damping matrix D_C is configured for critical damping [67]. In order to measure the human body configuration to shape the guidance behavior in Experiments 2 and 3, first, an Intel RealSense D435 RGB-D camera with OpenCV3 software [75] is used to detect the human keypoints. Color images are processed by the OpenPose library [76] that estimates human keypoints based on the BODY-25 model. Subsequently, the keypoint locations are deprojected using the aligned depth image; see Fig. 9 for an example of the model superposition over an RGB image. The keypoint positions are then fed to a simplified kinematics model of the human arm depicted in Fig. 10, to estimate the joint angles q_h and the Jacobian matrix $J_h(q_h)$. As the BODY-25 model considers only three keypoints for each arm (i.e., wrist, elbow, and shoulder), the arm keypoint locations can be used to identify the values of only three joints associated with arm abduction/adduction, arm flexion/extension, and elbow flexion/extension, depicted as joint 1, 2, and 4, respectively, in Fig. 10. Thus, due to this technical limitation, only three joints of the right arm kinematics are considered to build the body Jacobian matrix $J_h(q_h)$.

A. Experiment 1: Force-Driven Phase Evolution Control

In this set of experiments, the performance of the proposed force-driven phase-based motion generator introduced in Section III is evaluated. To achieve a consistent applied force on the robot, a 2.6 kg weight is attached to the robot's end-effector, as shown in Fig. 6. In all experiments, the constraint path is defined in 3-dimensional Cartesian space ($m = 3$) with a length of 70 cm. The path is vertical in Exp. 1.1, Exp. 1.3, and Exp. 1.4, and a circular arc in Exp. 1.2. The admittance parameters for Exp. 1.1, Exp. 1.2, and Exp. 1.3 are empirically configured to produce appropriate motion under the influence of the attached weight's gravity as follows:

$$m_\phi = 20, \quad d_\phi = 80, \quad c_\phi = 0 \quad (40)$$

and they are shaped according to (6)–(8) in Exp. 1.4.

1) *Experiment 1.1:* In this experiment, the path is uniformly encoded by phase as

$$x_d(\phi) = x_{d,0} + [0, 0, -0.7\phi]^T. \quad (41)$$

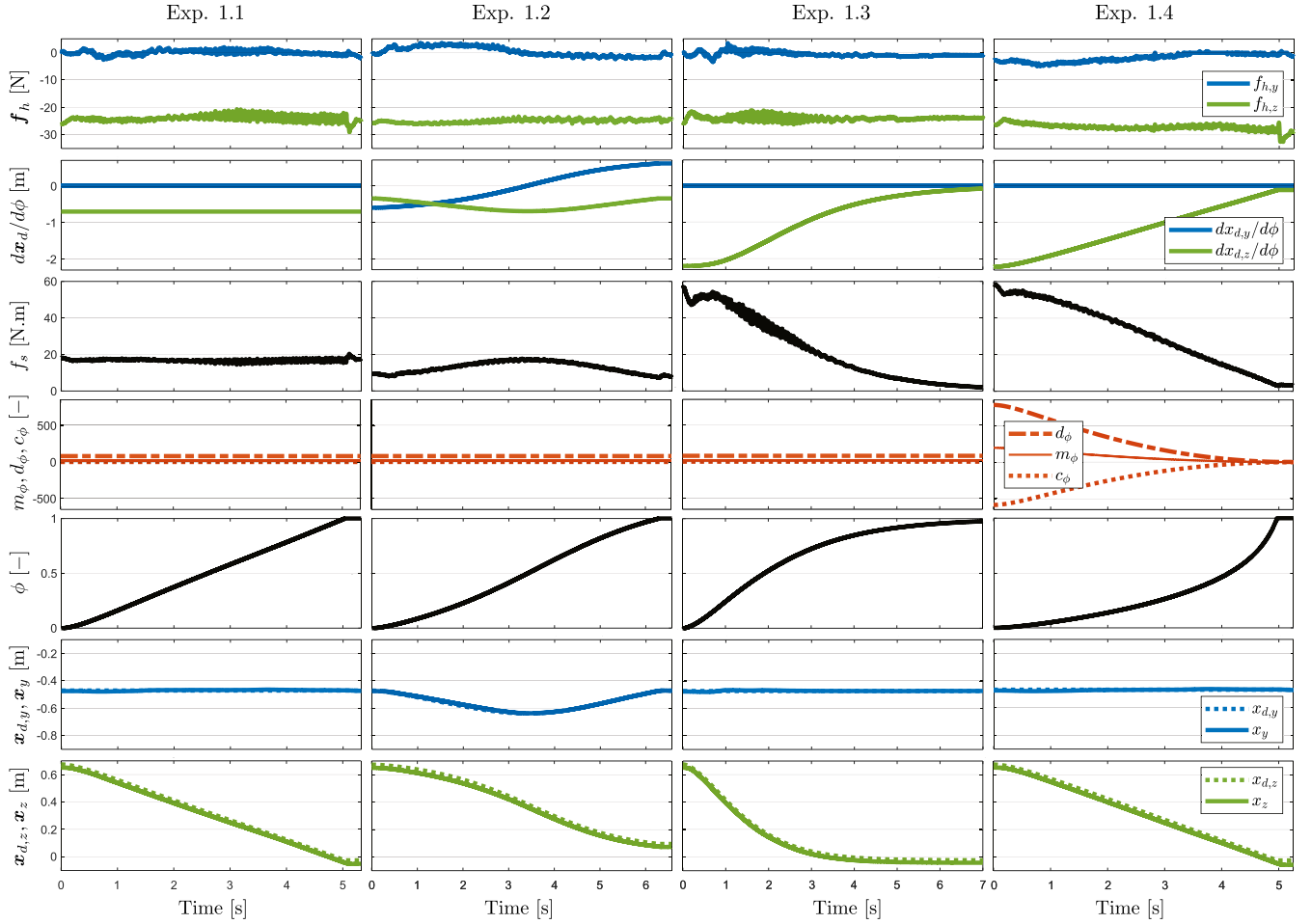


Fig. 7. Experiment 1 results; the attached weight applies a relatively constant force along the z -direction generating the phase trajectory. The resultant desired motions in Exp. 1.1 and Exp. 1.4 exhibit similarities due to the closely matched generalized admittance parameters m_a and d_a .

In (41), $\mathbf{x}_{d,0} \in \mathbb{R}^3$ defines the initial pose. As a result, the path direction vector $d\mathbf{x}_d(\phi)/d\phi$ for the encoded path (41) remains constant as follows:

$$d\mathbf{x}_d(\phi)/d\phi = [0, 0, -0.7]^T. \quad (42)$$

Thus, considering an approximate constant wrench $\mathbf{f}_h$ being applied to the robot, a constant steering force f_s guides the phase toward 1 according to (5). As shown in Fig. 6, the setpoint $\mathbf{x}_d$ proceeds throughout the path by the phase progress, and the impedance controller moves the robot accordingly.

2) *Experiment 1.2*: The path in this experiment is set to a 70 cm long arc equivalent to one-third of a full circle; see Fig. 6. It is again uniformly encoded as follows:

$$\mathbf{x}_d(\phi) = \mathbf{x}_{d,0} + 0.7 \frac{3}{2\pi} \begin{bmatrix} 0 \\ \sin(\frac{\pi}{6}) - \sin(\frac{2\pi\phi}{3} + \frac{\pi}{6}) \\ \cos(\frac{2\pi\phi}{3} + \frac{\pi}{6}) - \cos(\frac{\pi}{6}) \end{bmatrix} \quad (43)$$

with the direction vector

$$\frac{d\mathbf{x}_d(\phi)}{d\phi} = -0.7 \left[0, \cos\left(\frac{2\pi\phi}{3} + \frac{\pi}{6}\right), \sin\left(\frac{2\pi\phi}{3} + \frac{\pi}{6}\right) \right]^T. \quad (44)$$

Hence, while the path direction vector maintains a constant magnitude, it does not consistently align with the applied wrench $\mathbf{f}_h$. Consequently, the steering force f_s changes along the path, as depicted in Fig. 7. As a result, the evolution of the phase differs from the previous experiment. Specifically, at the midpoint of the path (i.e., for $\phi = 0.5$, due to the uniform encoding of the path), when the path direction vector aligns with gravity, f_s reaches its maximum value, thereby accelerating the progression of the phase.

3) *Experiment 1.3*: How the phase-based admittance law (5) generates the intended motion relies on the distribution of the variable phase throughout the encoded path, or in other words, how the path is encoded. To illustrate this, in contrast to the preceding scenarios, the path in this case—although akin to Exp. 1.1 in its exclusive alignment along the z -axis—is not uniformly encoded (see Fig. 6) as follows:

$$\mathbf{x}_d(\phi) = \mathbf{x}_{d,0} + \left[0, 0, -0.7 \frac{1 - e^{-3\phi}}{1 - e^{-3}} \right]^T \quad (45)$$

and as a result, the path direction vector becomes

$$d\mathbf{x}_d(\phi)/d\phi = \frac{-2.1}{1 - e^{-3}} [0, 0, e^{-3\phi}]^T \quad (46)$$

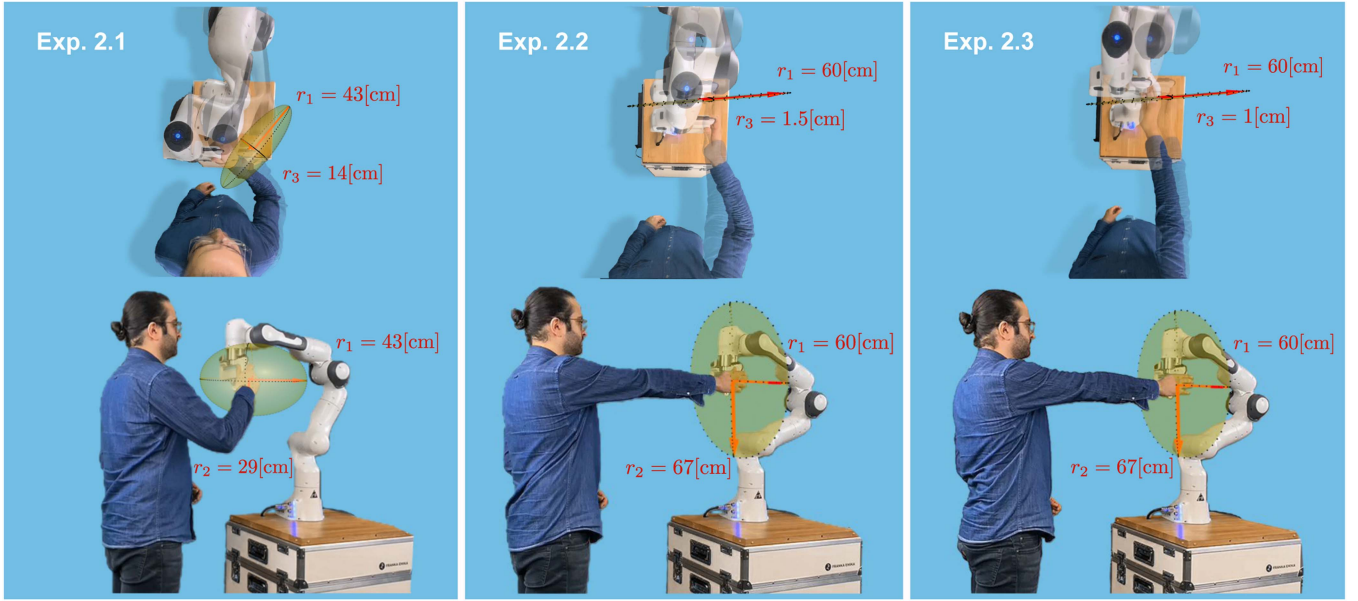


Fig. 8. Experiment 2 scenarios; lower figures display initial configurations, while upper figures depict motion along the 10 cm paths. The path in Exp. 2.1 and Exp. 2.3 is to the left of the operator (along the robot's x -axis), while in Exp. 2.2, it is toward the operator (along the robot's y -axis). Manipulability ellipsoids and their principal radii are shown for operator arm initial configurations. To enhance detail, ellipsoids are presented at half their actual size.

which has no constant norm anymore. At the path's outset ($\phi = 0$), the tangent vector norm is about $22\times$ that at the path's end ($\phi = 1$). This leads to higher initial values of f_s , resulting in pronounced accelerations in both ϕ and $\mathbf{x}_d(t)$ compared to Experiment 1.1, as shown in Fig. 7. As the phase approaches 1, the steering force diminishes, significantly slowing down the phase progression.

4) *Experiment 1.4:* In Section III, a more intuitive approach to parameter tuning is suggested here, as opposed to adjusting the phase-based parameters m_ϕ , d_ϕ , and c_ϕ directly. This approach involves setting the generalized parameters m_a and d_a first and then deriving the phase-based parameters using (6)–(8). In this experiment, the same encoded path as in Experiment 1.3 is used but by considering desired admittance mass $m_a = 40$ kg and damping coefficient $d_a = 160$ Ns/m. As shown in Fig. 7, the phase-based parameters evolve along the path due to changes in the path direction vector $d\mathbf{x}_d(\phi)/d\phi$ in (46). At the path's outset, $d\mathbf{x}_d(\phi)/d\phi$ has greater magnitude, leading to larger phase-based parameters. However, as ϕ approaches 1, the vector's magnitude decreases, causing reductions in m_ϕ , d_ϕ , and c_ϕ .

These admittance parameters are shaped such that the phase progress $\phi(t)$ yields a desired motion $\mathbf{x}_d(t)$ akin to motion resulting from applied wrench $\mathbf{f}_h$ on a 40 kg mass with viscous friction coefficient 160 Ns/m. In other words, by utilizing generalized admittance parameters m_a and d_a and adjusting phase-based parameters accordingly, desired motion generation (5) becomes independent of the phase-distribution policy. Fig. 7 illustrates that generated desired motions in Experiment 1.1 and Experiment 1.4 are nearly identical despite differing path encodings. This similarity arises because the phase-based parameters in Experiment 1.1 are comparable to those obtainable by using the path direction vector (42), along with generalized

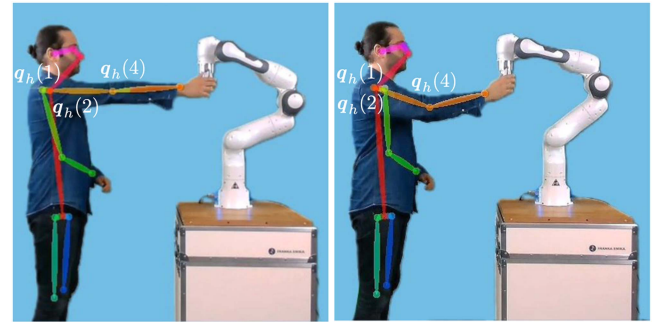


Fig. 9. Body configuration at the start (left) and end (right) of Exp. 2.2. Joint angles are estimated using the detected body keypoints' positions.

parameters $m_a = 40$ and $d_a = 160$ in (6)–(8), as follows:

$$m_\phi = 19.6, \quad d_\phi = 78.4, \quad c_\phi = 0. \quad (47)$$

These values are almost equivalent to the ones in (40).

B. Experiment 2: Human-Aware Motion Guidance Adaptation

This set of experiments aims to explore the behavior of the guidance law with the incorporation of parameter adaptation. The adaptation process is based on the exemplary policy outlined in Section IV. Here, a human operator is supposed to guide the robot through a 10 cm predefined uniformly encoded path by applying wrenches on the robot end-effector; see Fig. 8. As previously mentioned, the human body configuration is determined by detecting and locating the keypoints (shoulder, elbow, and wrist) using the OpenPose software [76] and a RealSense RGB-D camera, as shown in Fig. 9. Using the detected keypoint positions along with the 7-DoF kinematics model of the human

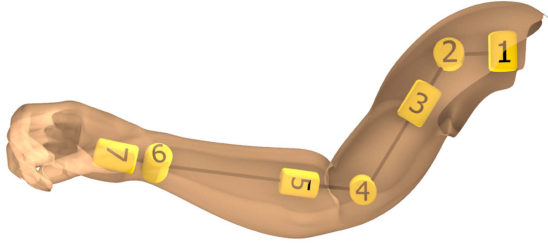


Fig. 10. Arm kinematic structure introduced in [77]. In the experiments, only the 1st, 2nd, and 4th joints are used to calculate the Jacobian matrix.

arm from [77], three joint angles $q_h(1)$, $q_h(2)$, and $q_h(4)$ are identified (see Fig. 10). These angles are used to calculate the Jacobian matrix $J_h(q_h)$. Consequently, the manipulability ellipsoid is continuously recognized, and the metric l is determined by projecting the principal radii of the ellipsoid onto the desired path, as explained in Section IV. Across all experiments, the minimum and maximum values for the principal radii lengths are defined as $r_{\min} = 0$ and $r_{\max} = 70$ cm, respectively. The admittance mass m_a is shaped based on (17), using minimum and maximum values of $m_{a,\min} = 1$ kg and $m_{a,\max} = 80$ kg, respectively. Finally, the damping coefficient d_a is consistently set to half of m_a throughout all experiments. These experiments showcase how online adaptation of admittance parameters operates in practice. They aim to observe the impact of changes in the admittance parameters on guidance behavior for the same task, particularly in terms of the human input required.

1) *Experiment 2.1:* In the first experiment, the operator arm's initial configuration is approximately $q_h \approx [75, 45, 97]^T$ degrees. This configuration leads to the manipulability ellipsoid illustrated in Fig. 8. Projecting the principal radii of this ellipsoid onto the encoded path—a straight line to the operator's left side—yields $l \approx 26$ cm, corresponding to an admittance mass of approximately $m_a \approx 50$ kg. The operator guides the robot by applying the wrench f_h , resulting in the generation of the steering force f_s , visible in Fig. 11.

2) *Experiment 2.2:* In this experiment, the human operator's arm starts in an extended position near a singularity, resulting in an extremely elongated manipulability ellipsoid, depicted in Fig. 8. The constraint path aligns as a straight line following the human arm, requiring the operator to pull the robot toward themselves; see Fig. 9. At $\phi = 0$, projecting the ellipsoid radii onto the encoded path yields a minimal metric value, $l \approx 4$ cm. This metric corresponds to a high admittance mass of around $m_a \approx 76$ kg. As elucidated in Section IV, such an arm's configuration results in low joint torques for the necessary guiding wrench. Conversely, a relatively high joint velocity may be necessary for swift movement along the path. Thus, a high admittance mass is preferable in this scenario. As illustrated in Fig. 11, achieving the desired motion along the path requires notably elevated steering forces. Despite these considerable forces, completing the 10 cm path takes the longest duration, approximately $t \approx 1.8$ s.

3) *Experiment 2.3:* Though the human arm configuration resembles the previous case, the constraint path in this experiment closely follows the ellipsoid's longer radius (see Fig. 8). Consequently, the initial projection norm $l \approx 67$ cm is in close

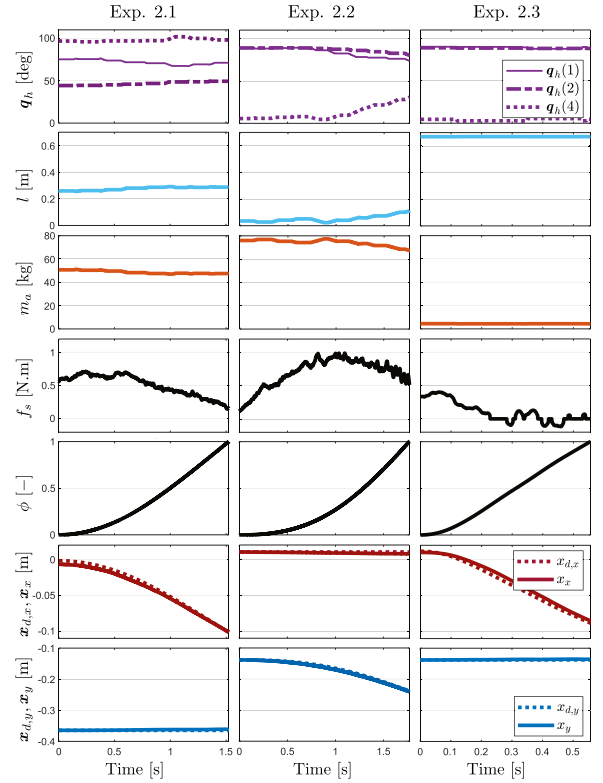


Fig. 11. Experiment 2 results; the mass m_a continuously adapts based on the norm l . The average force magnitudes applied in the path direction are -5 N, -7 N, and -1 N for Exp. 2.1, Exp. 2.2, and Exp. 2.3, respectively.

proximity to $r_{\max}$. This proximity results in a relatively low admittance mass of $m_a \approx 4$ kg. While the configuration is similar to the previous case, the altered guidance direction may require significant joint torques to achieve the same magnitude as the guiding wrench. Hence, the reduced admittance mass in this experiment serves to diminish the necessary magnitude of the guiding wrench. Unlike in Experiment 2.1, as evident in Fig. 11, the joint motions required for the same Cartesian motion are comparatively minor. This observation aligns with the manipulability analysis outlined in Section IV. In fact, even the swift Cartesian motion ($\phi = 1$ at $t = 0.55$ s) does not result in high joint velocities.

C. Experiment 3: The Effect of Energy Tank Augmentation

Previous experiments demonstrate how applying a wrench to the robot can guide the setpoint along a constraint path and how the guidance response varies with different m_a values. Moreover, the adaptability of m_a is shown according to the human body configuration for a desired guidance behavior. Subsequent experiments reveal potential issues with shaping m_a , leading to undesired energy generation that violates passivity. A solution involving a virtual energy tank is applied to curb this by constraining the rate of admittance mass change, $\dot{m}_a$, ensuring it remains below the permissible μ as defined in (33). In this scenario, an operator guides the robot along a uniformly encoded path with changing arm configuration; see Fig. 13. The guidance adaptation policy mirrors the earlier experiment.

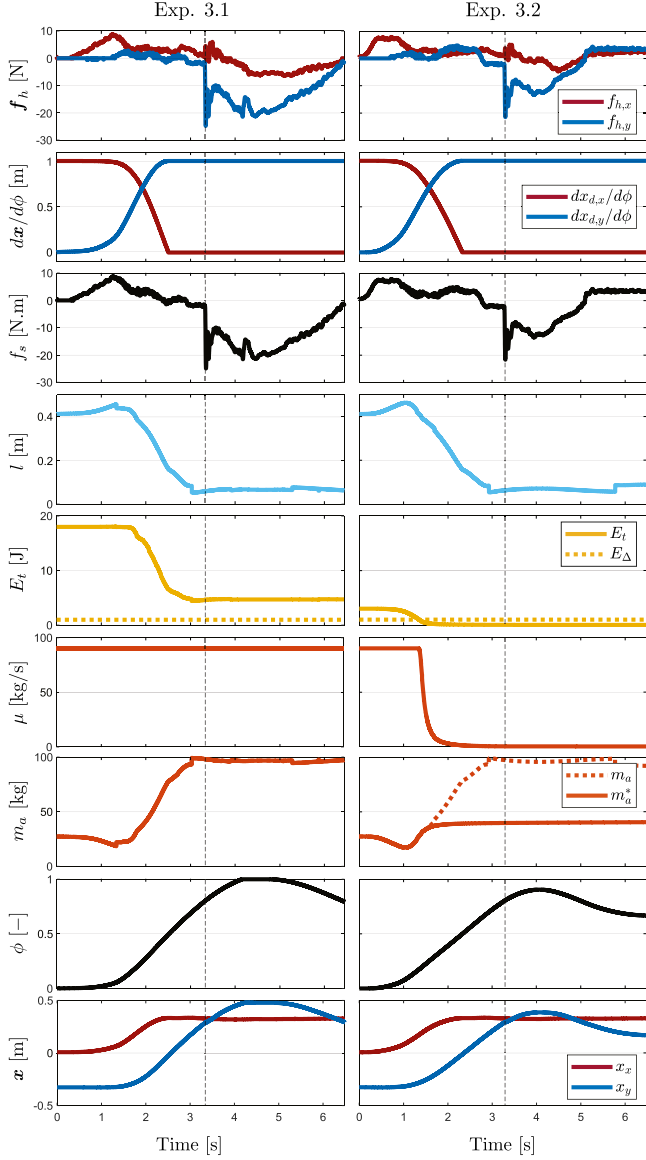


Fig. 12. Experiment 3 results; unlike Exp. 3.1, tank's low initial energy in Exp. 3.2 prevents the admittance mass from increasing. Consequently, m_a^* remains below m_a . The dashed vertical lines mark the collision time.

The minimum and maximum manipulability ellipsoid radii are set to $r_{\min} = 5$ cm and $r_{\max} = 50$ cm, with corresponding admittance mass $m_{a,\min} = 10$ kg and $m_{a,\max} = 100$ kg. Damping coefficient is $d_a = m_a/4$. As the operator configuration alters, l decreases, and m_a increases. At $\phi = 0.8$ along the path, a collision occurs with a pendulum bearing a 5 kg weight, depicted in Fig. 13. Two experiments are presented: one with high initial tank energy, insignificantly affecting performance, and another with low initial energy, limiting mass variation. Both cases utilize an energy threshold $E_\Delta = 1$ J. The posttank admittance mass is denoted as m_a^* , accompanied by its virtual kinetic energy E_{k_a} defined as

$$E_{k_a} = \frac{1}{2} m_a^* \dot{x}_d^T \dot{x}_d. \quad (48)$$

The impact of the tank's initial energy level on the virtual kinetic energy E_{k_a} and, consequently, the behavior after collision is illustrated in Fig. 13 for each experiment.

1) *Experiment 3.1:* As depicted in Fig. 12, the initial three seconds showcase the human operator exerting the interaction wrench f_h on the robot. This action produces the steering force f_s , propelling the robot along the path. The projection norm l and consequently the admittance mass m_a vary due to changes in the operator's arm configuration. Despite the high initial energy in the tank ($E_t(t=0) = 18$ J), the energy level diminishes as m_a increases. However, since the tank's energy never reaches the E_Δ threshold, the tank output μ remains constant. Thus, the admittance mass can freely adjusted using (17). These admittance parameters, m_a and d_a , influence the phase-based parameters in the guidance law (5), ultimately giving rise to the desired motion $x_d(\phi)$, as shown in Fig. 12.

At around $t \approx 3$ s, the operator withdraws their hand, and at approximately $t \approx 3.3$ s, the robot collides with the pendulum. The impact force abruptly opposes the steering force, countering the phase progression. By this moment, the low projection norm l caused by the extended arm configuration leads to m_a reaching its peak value. Consequently, a substantial virtual momentum arises, generating high virtual kinetic energy ($E_{k_a} = 5.5$ J). Therefore, during the collision, the opposing force f_s is insufficient to halt phase advancement. As the phase hits 1, the desired trajectory halts, and due to the pendulum tilting, gravitational force strengthens, driving the phase in reverse. Subsequently, the robot traces the desired trajectory back toward the operator.

2) *Experiment 3.2:* The initial energy within the virtual tank for this experiment is set at 3 J. Consequently, the rise in admittance mass m_a gradually consumes the tank's energy until it reaches the lower threshold E_Δ . At this juncture, the tank's output μ decreases to zero, and the ability to shape m_a ceases. This leads to a distinctive outcome compared to the previous experiment, where the virtual kinetic energy remains steady at $E_{k_a} = 1.5$ J. Subsequently, the robot's postimpact behavior differs. The impact force from the pendulum, while weaker than in Experiment 3.1, proves sufficient to arrest and reverse the robot's motion prior to completing the path, specifically at $\phi = 0.9$. The findings clearly illustrate that the drained tank energy during the rise of admittance mass in Experiment 3.1 amplifies the virtual kinetic energy of the motion generator. This energy is then expended to fully elevate the pendulum post-impact. On the other hand, the constrained tank energy in Experiment 3.2 limits the generated virtual kinetic energy, insufficient to entirely raise the pendulum.

D. Experiment 4: Phase-Based Admittance Control vs. Virtual Mechanism

In this set of experiments, the performance of the proposed phase-based admittance law is evaluated compared to the well-established approach of the virtual mechanism, initially introduced in [16]. It is a common method for generating virtual guides and shares some similarities with the proposed guidance law. For instance, this approach employs a phase-encoded path

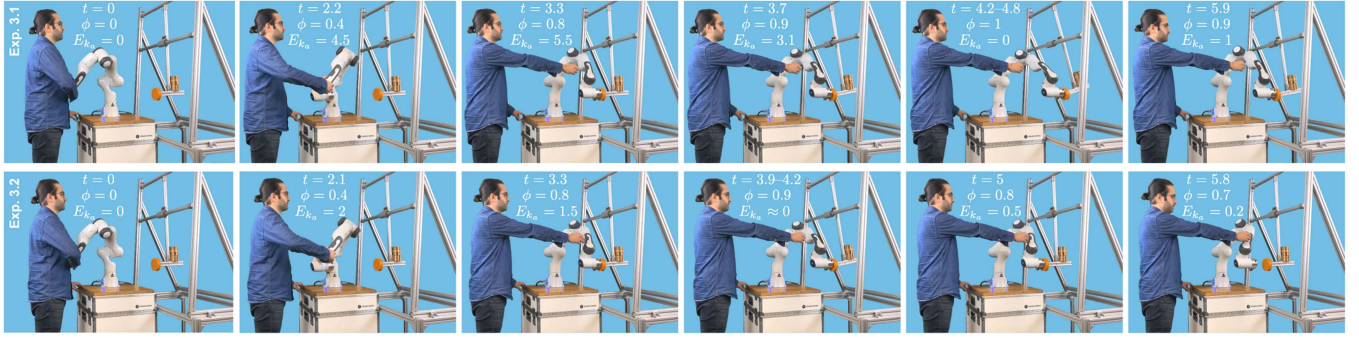


Fig. 13. Experiment 3 scenarios; the operator guides the robot through the path, and the robot collides with a pendulum. The generated high virtual kinetic energy in Exp. 3.1 pushes the pendulum to the maximum until $\phi = 1$. In contrast, the augmented virtual tank in Exp. 3.2 prevents the virtual kinetic energy from growing, and as a result, the impact force is able to stop the desired motion. Time t and energy E_{k_a} are indicated in seconds and joules, respectively.

with initial and end points associated with phase values of 0 and 1, respectively. A virtual spring-damper system is assigned between the setpoint and the robot that applies the following wrench to the robot:

$$\mathbf{f}_{vm} = \mathbf{K}_{vm}(\mathbf{x}_d - \mathbf{x}) + \mathbf{D}_{vm}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}). \quad (49)$$

Here, $\mathbf{K}_{vm}, \mathbf{D}_{vm} \in \mathbb{R}^{m \times m}$ represent the stiffness and damping of the system, respectively. To determine the desired trajectory for the spring-damper system, the following condition is taken into account:⁹

$$\mathbf{f}_{vm}^T \frac{d\mathbf{x}_d(\phi)}{d\phi} = 0. \quad (50)$$

By inserting (49) into (50) and considering $\dot{\mathbf{x}}_d = \dot{\phi} d\mathbf{x}_d/d\phi$, the phase dynamics become [18]

$$\dot{\phi} = \frac{d\mathbf{x}_d^T/d\phi (\mathbf{K}_{vm}(\mathbf{x} - \mathbf{x}_d) + \mathbf{D}_{vm}\dot{\mathbf{x}})}{d\mathbf{x}_d^T/d\phi \mathbf{D}_{vm} d\mathbf{x}_d/d\phi}. \quad (51)$$

By obtaining the phase trajectory $\phi(t)$ from (51), and inserting it into the encoded path $\mathbf{x}_d(\phi)$, the desired trajectory $\mathbf{x}_d(t)$ is consistently determined for the derivation of the input torque $\boldsymbol{\tau}_{vm} = \mathbf{J}^T(\mathbf{q})\mathbf{f}_{vm}$ from (50).

When comparing (51) with (4)–(5), a key difference between the virtual mechanism approach and the proposed approach becomes evident. In the virtual mechanism approach, it is not possible to adapt the parameters of the phase trajectory generator independently of motion control. Consequently, achieving a faster phase speed requires an increase in the wrench applied by the human. In contrast, the proposed phase-based admittance law (5) allows the parameters to be adjusted independently, enabling the phase progression speed to be shaped without altering the applied wrench. This distinction is explored in the current set of experiments.

To compare the performance of the two approaches, four experiments are conducted (see Fig. 14). A uniformly phase-encoded horizontal line with a length of 20 cm along the robot's x -direction is assigned as the desired path. The end-effector's rotation is fixed, allowing the robot to move only

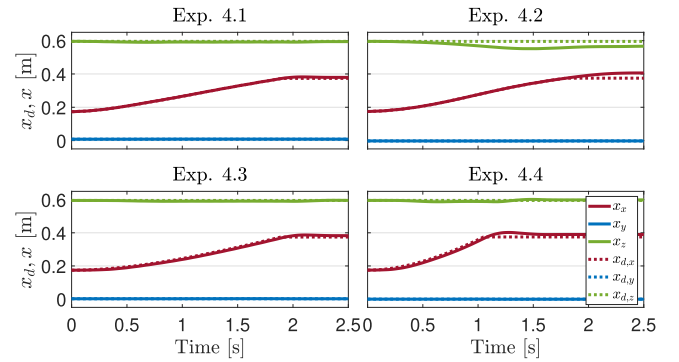


Fig. 14. Experiment 4 results; unlike virtual mechanism (Exp. 4.1, 4.2), the phase-based admittance law allows for modifying guidance characteristics, such as enabling faster motion with the same applied wrench (Exp. 4.3, 4.4). In Exp. 4.2, the robot deviates from the constraint path (specifically in the z -direction) due to the lower stiffness and damping values.

translationally. The objective of these experiments is to determine whether the characteristics of each guidance law can be shaped by adapting the parameters of each approach, without compromising the geometric constraint.

1) *Experiment 4.1:* In the first experiment, the virtual mechanism approach is utilized with diagonal stiffness and damping matrices having elements $\{1000, 1000, 1000\}$ N/m and $\{40, 40, 40\}$ Ns/m, respectively. To simulate a consistent human-applied wrench to the robot, a 5 N feedforward wrench in the direction of the desired path is superimposed on the virtual mechanism wrench $\mathbf{f}_{vm}$ and applied to the robot. As shown in Fig. 14, this wrench moves the robot along the encoded path in approximately 2 s, and due to the relatively large stiffness and damping values, the robot remains on the path.

2) *Experiment 4.2:* The same setup as in Experiment 4.1 is utilized here, but with stiffness and damping matrices having elements $\{100, 100, 100\}$ N/m and $\{4, 4, 4\}$ Ns/m. As shown in Fig. 14, during motion, the robot slightly deviates from the path, specifically in the z -direction. This deviation occurs because the small stiffness and damping matrices are insufficient to compensate for disturbance wrenches perpendicular to the path, such as those resulting from joint friction. It is noteworthy that the time required to traverse the path with the applied 5 N wrench remains almost the same as in the previous experiment.

⁹This condition stems from the assumption that no effort is transferred to the setpoint itself, which can be represented as a massless particle moving along the path [18].

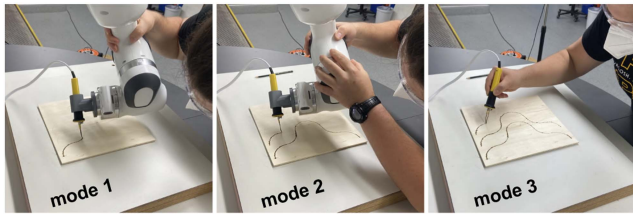


Fig. 15. Experiment 5 setup; pyrography on a wood plate with three different modes; the operator should create a consistent line along the drawn paths.

3) *Experiment 4.3:* In this experiment, the proposed phase-based admittance law (4)–(5) is employed. The admittance mass m_a and damping d_a are set to 26 kg and 26 Ns/m, respectively. To facilitate better comparability, the same spring-damper system (49) is used as the robot's motion controller, with diagonal stiffness and damping matrices having elements $\{1000, 1000, 1000\}$ N/m, $\{40, 40, 40\}$ Ns/m, respectively. The guiding wrench f_h is manually set to 5 N along the robot's x -axis. Fig. 14 depicts how with the employed admittance parameters, the same amount of time as in the previous experiments is needed to traverse through the path with the applied 5 N wrench.

4) *Experiment 4.4:* This experiment employs the same setup as in Experiment 4.3, yet with smaller admittance mass and damping values 10 kg and 10 Ns/m, respectively. As can be seen in Fig. 14, lowering the admittance parameters results in a shorter time length to reach the path's end-point using the same 5 N applied wrench as in the previous experiments. It can be observed that, in contrast to Experiment 4.2, the robot's deviations from the path are negligible while moving along it.

This set of experiments highlights a notable advantage of the proposed guidance law over the well-established virtual mechanisms. The modularity of the phase-based admittance policy allows for adjustments of the guidance law's characteristics, such as making it easier or more difficult to move the robot, without significantly compromising control performance, such as maintaining the robot on a constraint path. Consequently, the guidance behavior can be easily shaped according to the task requirements, such as respecting the operator's manipulability, as shown in Section IV. This adaptability is absent in virtual mechanisms, where guidance dynamics are tightly intertwined with motion control.

E. Experiment 5: Usability Study in Pyrography

Compared to previous experiments showcasing the proposed methods' functionality, this experiment focuses on evaluating the practicality of the path-constrained motion guidance law. Here, a pyrography task where moving a heated metal tip on a wooden surface creates a line mark is examined (see Fig. 15). Line thickness is controlled by adjusting metal pressure and movement speed, demanding both precision (staying on the line) and robustness (coping with wood surface or metal temperature changes). The objective is a consistent mark along a predefined line (see Fig. 15). A disturbance is introduced by reducing metal temperature midway, potentially causing line inconsistency if the burner speed is not adjusted. Fig. 16 depicts the result of a



Fig. 16. Experiment 5 results. Automated pyrography with an impedance-controlled robot along a predefined path at a constant speed. The wood burner temperature intentionally drops by a quarter in the middle, resulting in a fainter marked line thereafter.

robot moving the wood burner at a constant speed on various wood types; a fainter mark on the latter half underlines the need to address the disturbance. As advanced sensors may be absent, human involvement is crucial to visually assess the quality and adapt the burner's pace.

To assess the suggested motion guidance policy, a user study¹⁰ was devised. Twenty volunteers were tasked with creating a uniform line along drawn paths on 20 wood plates using three modes (see Fig. 15).

- 1) Mode 1: The robot holds the wood burner, and the human operator guides the motion via the proposed guidance law (i.e., the phase-based motion generator + the impedance control + gravity compensation).
- 2) Mode 2: The robot holds the wood burner, and keeps its orientation constant, allowing the human to freely adjust the wood burner's position. (i.e., impedance control with zero translational stiffness + gravity compensation).
- 3) Mode 3: The human operator holds the wood burner in hand without a robot.

In mode 1, the desired path is uniformly encoded, and the stiffness of the impedance controller orthogonal to the plate surface is set to zero. This enables the operator to easily lift the robot's end-effector, removing the wood burner from the plate (see Fig. 15). Notably, any online parameter adaptation is excluded from the guidance law to maintain the focus of the experiments on evaluating the performance of the phase-based guidance policy in its basic form. Nevertheless, instead, after a few preliminary trials, the parameters m_a and d_a are fixed based on each user's preference. Safetywise, the human is instructed to keep their hand away from both the wood burner tip and the wood surface. Importantly, unlike Experiment 4, the intention behind this user study is not to compare the proposed guidance algorithm with related literature. Instead, the focus is on gaining firsthand insights into the proposed approach when used alongside the two basic methods. In this context, it is already expected that the proposed approach (mode 1) will outperform the other methods, as it is specifically tailored to the task, particularly with the impedance control in place. Nonetheless, the primary aim is to analyze participants' experiences when using such an assistive system for a path-constrained motion-following task.

¹⁰The study was conducted with approval from the Ethics Commission of the Technical University of Munich (review number 744/21 S-SR).



Fig. 17. Experiment 5 results. Twenty participants utilized three distinct modes to produce uniform lines on twenty wooden plates. The lines on the left, center, and right of each plate correspond to mode 1, mode 2, and mode 3, respectively.

A total of 20 participants took part, including an equal split of ten males and ten females, with ages ranging from 22 to 61 and an average of 31.3 years. Initially, each participant's table height was adjusted, and they were given a few minutes to acquaint themselves with pyrography. Subsequently, the experiment and questionnaire were explained. Additionally, they had a brief period before each mode to practice and, especially in mode 1, set their personalized, comfortable admittance parameters. The order of modes was randomized for each user to prevent potential questionnaire response bias. In the questionnaire, participants were first queried about their experience in physical interaction with robots, yielding an average response of 2.45 on a scale of 1 (none) to 5 (very experienced). They were also asked about their prior involvement in crafts or pyrography-related activities, resulting in an average response of 2.9. After this, participants engaged in the task using one of the three modes and promptly responded to the following mode-specific questions using a Likert scale (ranging from strongly disagree to strongly agree).

- 1) Comfort: I felt comfortable during the task.
- 2) Freedom: My body posture was unconstrained, and I was free to change it during the task.
- 3) Control: I had complete control over the task performance.
- 4) Ease: It was easy to fulfill the task requirement of staying on the path.
- 5) Safety: I felt safe during the task.

Finally, on a scale of 1 to 5, with 1 being the worst and 5 being the best evaluation score, the participants replied to the two following questions.

- 1) Efficacy: How would you rate your ability to deliver a good result with the approach (given your current training level)?
- 2) Overall: What is your overall evaluation of the approach?

Fig. 17 illustrates the resulting wood plate shapes, while Fig. 18 presents the questionnaire outcomes. Mode 1 garnered the highest scores across most questions. On average, participants found mode 1 notably easier and more effective in achieving the desired outcome. Specifically, among participants with handicraft expertise below 3, the *comfort* question had an average response of 4.7 for mode 1, compared to 3.4 and 2.7 for mode 3 and mode 2, respectively. This subgroup is also considered mode 3 to be the least safe, with an average rating of 3.9. Intriguingly, participants with handicraft expertise above 3 still awarded the highest average score for the *efficacy* question to mode 1 (4.4), followed by mode 3 (4.1) and mode 2 (3). Regarding the *freedom* question, mode 3 garnered the highest overall average response. An interesting observation emerged

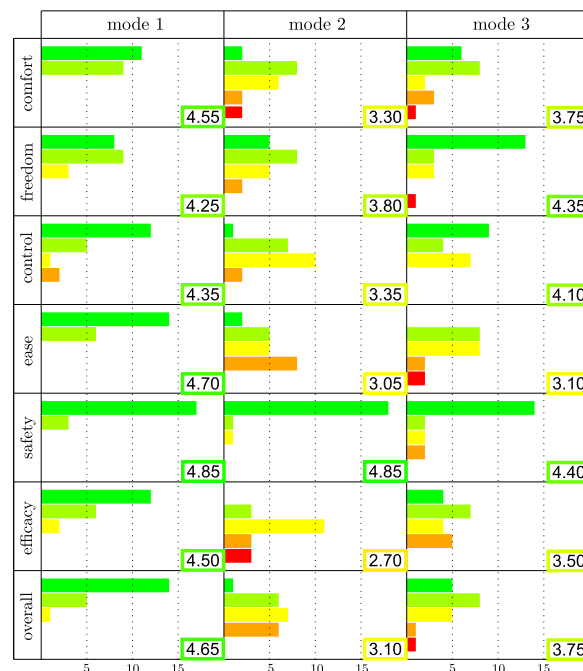


Fig. 18. Experiment 5 questionnaire results; in each box, bars of varying colors (from green to red, top to bottom) represent votes for each question and mode. They encode responses from *strongly agree* to *strongly disagree* for the initial five questions and scores 5 to 1 for the last two questions. Bar lengths correspond to the number of votes, and the average vote is displayed in the bottom-right corner of each box.

from participants with a robot interaction experience level of 1 or 2, who assigned a 4.8 average score for the *control* question to mode 1, compared to 3.6 for mode 2 and 3.9 for mode 3. This suggests that, relative to the overall ratings, they had a more favorable experience when the robot was engaged. Notably, their mean response for the question regarding overall experience was 4.9 for mode 1. Lastly, time was another factor under consideration. Mode 1 proved the fastest for 14 participants, whereas mode 3 was the fastest for only 4 participants. Conversely, mode 3 was the slowest for 10 participants, followed by mode 2 for 7 participants. Please note that in addition to time, other quantitative metrics, such as accuracy checks of the generated curves or physical interaction power, could have been used to evaluate the three modes. As shown in Fig. 17, mode 1 generally produced more consistent marks. Given that it also had the lowest average time, even comparable outcomes suggest that mode 1 is more effective. A more detailed analysis, including these additional metrics, is planned for future research.

VII. CONCLUSION

In the problem addressed in this article, the human was tasked with guiding strictly constrained robot movements, where the generated motion must consistently adhere to a predefined path. To tackle this, a novel admittance law was introduced—employing an auxiliary variable called phase—which mapped the human-applied wrench to the motion along the constraint path. This phase-based haptic guidance law was then demonstrated to be generalizable using standard physically interpretable parameters. Additionally, it was shown how these generalized parameters can be adjusted based on the human's preferences. The tuning process was dynamically conducted during the guidance task, leveraging the human body configuration-based manipulability assessment. A passivity analysis was derived revealing that such adaptation could introduce instability. To mitigate this, a solution was developed by deriving a virtual energy tank. The practical aspects of the proposed guidance approach were explored through various sets of experiments, each highlighting different facets. Specifically, the focus had been on the choice of the path encoding method and guidance parameters, the process of guidance adaptation utilizing a vision system and the manipulability concept, and finally, the alterations in interaction behavior due to different parametrizations of the virtual tank. Next, a set of experiments was conducted to compare the performance of the proposed guidance law with respect to the well-established approach of virtual mechanisms. Lastly, a usability study involving a relevant task, i.e., pyrography, was conducted to evaluate how different users perceive and respond to the proposed guidance policy.

The proposed phase-based guidance law was modular and applicable to any motion-controlled robot capable of sensing the human-applied wrench. However, for the stability analysis, it was assumed that the guided robot was equipped with an impedance controller. Therefore, if other motion controllers were used, the stability analysis needs to be updated. The proposed guidance adaptation objective served as an illustrative example, and any alternative approach can be employed without compromising the stability analysis procedure. In essence, the augmented virtual tank was introduced to address potential passivity-violating powers stemming from the adaptation of the admittance law, regardless of the specific adaptation goal. Consequently, in cases where no adaptation occurred, such as in Experiment 5, the system retained passivity without necessitating a virtual energy tank. It is important to note that this assumption holds true only when a finite amount of energy can be attributed to the guiding human.

As demonstrated in Fig. 1, the proposed guidance policy can be effectively applied not only in human–robot comanipulation scenarios, such as cases a and c in Fig. 1, and the analyzed pyrography use case, but also in other contexts. For example, in teleoperation scenarios, such as case b in Fig. 1, where it was necessary to constrain the motion of either the leader or the follower, or in rehabilitation scenarios, like case d in Fig. 1, where it was crucial for the patient to follow specific therapeutic paths, this approach was equally applicable.

One limitation of the proposed phase-based admittance law compared to similar approaches, such as the virtual mechanism, was its dependence on the observation of the guiding wrench,

which posed at least two main challenges. First, any error in the estimation of the applied wrench can potentially lead to an incorrect generation of the desired trajectory. Nevertheless, due to the modularity of the approach, the path constraint would not be violated as long as the motion controller remains active. The second challenge was distinguishing between intentional and unintentional wrenches. This distinction was particularly relevant when other environmental wrenches were present. This included the gravity of an unmodeled object being held by the robot (e.g., during assembly tasks), or friction and other contact wrenches, when the task involved significant physical interaction with the environment (e.g., sawing). Experiment 3 illustrated this issue, showing that the guidance law treated human-applied and pendulum-applied wrenches equally. To address these scenarios, one potential solution was to use a separate sensor dedicated to detecting human-applied wrenches. Alternatively, incorporating the dynamics model of the object or task could help to differentiate the wrenches more accurately. These features could be integrated into the guidance policy and are planned for future work.

APPENDIX A

DERIVATION OF THE ADMITTANCE PHASE-BASED PARAMETERS

The goal is to establish a mass-damper admittance property between the contact wrench $\mathbf{f}_h$ and the desired motion $\mathbf{x}_d(t)$ when $\mathbf{f}_h$ is aligned with the path tangent vector $d\mathbf{x}_d/d\phi$. The admittance relation then becomes

$$m_a \ddot{\mathbf{x}}_d + d_a \dot{\mathbf{x}}_d = \mathbf{f}_h. \quad (52)$$

Considering the following chain rules:

$$\dot{\mathbf{x}}_d = \dot{\phi} \frac{d\mathbf{x}_d}{d\phi}, \quad \ddot{\mathbf{x}}_d = \dot{\phi}^2 \frac{d^2\mathbf{x}_d}{d\phi^2} + \ddot{\phi} \frac{d\mathbf{x}_d}{d\phi} \quad (53)$$

the relation (52) can be rewritten in transpose form as

$$m_a \ddot{\phi} \frac{d\mathbf{x}_d^T}{d\phi} + m_a \dot{\phi}^2 \frac{d^2\mathbf{x}_d^T}{d\phi^2} + d_a \dot{\phi} \frac{d\mathbf{x}_d^T}{d\phi} = \mathbf{f}_h^T. \quad (54)$$

Multiplying $d\mathbf{x}_d(\phi)/d\phi$ from the right side of (54) yields

$$\underbrace{\ddot{\phi} m_a \frac{d\mathbf{x}_d^T}{d\phi} \frac{d\mathbf{x}_d}{d\phi}}_{m_\phi} + \underbrace{\dot{\phi}^2 m_a \frac{d^2\mathbf{x}_d^T}{d\phi^2} \frac{d\mathbf{x}_d}{d\phi}}_{c_\phi} + \underbrace{\dot{\phi} d_a \frac{d\mathbf{x}_d^T}{d\phi} \frac{d\mathbf{x}_d}{d\phi}}_{d_\phi} = \underbrace{\mathbf{f}_h^T \frac{d\mathbf{x}_d}{d\phi}}_{f_s}. \quad (55)$$

APPENDIX B

DYNAMIC MOVEMENT PRIMITIVE VERSUS PHASE-BASED GUIDANCE LAW

The standard formulation of a discrete dynamic movement primitive (DMP) is represented as [21]

$$\dot{\mathbf{z}}_z = \Omega (\alpha_z (\beta_z (\mathbf{g}_z - \mathbf{x}_{\text{DMP}}) - \mathbf{z}_z) + \mathbf{f}_z(\phi)) \quad (56)$$

where $\mathbf{x}_{\text{DMP}} \in \mathbb{R}^m$ denotes the system state representing the encoded motion trajectory, $\mathbf{g}_z \in \mathbb{R}^m$ is the goal state, and $\alpha_z, \beta_z \in \mathbb{R}_{\geq 0}$ are DMP gains. Additionally, $\mathbf{f}_z(\phi) \in \mathbb{R}^m$ is the nonlinear forcing term and $\Omega \in \mathbb{R}$ is the phase velocity determining how fast the motion progresses via

$$\Omega = \dot{\phi}. \quad (57)$$

Finally, $\mathbf{z}_z \in \mathbb{R}^m$ corresponds with the DMP speed such that

$$\dot{\mathbf{x}}_{\text{DMP}} = \Omega \mathbf{z}_z. \quad (58)$$

To facilitate a comparison without altering the derivation, the variable conventions used in (56) are adjusted as follows:

$$\Omega := \frac{1}{\sqrt{m_a}}, \quad \mathbf{x}_{\text{DMP}} := \mathbf{x}_d \quad (59)$$

$$\alpha_z := \frac{d_a}{\sqrt{m_a}}, \quad \beta_z := \frac{k_a \sqrt{m_a}}{d_a}. \quad (60)$$

Thus, the DMP formulation (56) transforms into

$$m_a \ddot{\mathbf{x}}_d + d_a \dot{\mathbf{x}}_d = k_a (\mathbf{g}_z - \mathbf{x}_d) + \mathbf{f}_z(\phi). \quad (61)$$

Comparing (61) with (9), the key difference between DMPs and the proposed guidance law becomes evident. In a DMP, the desired trajectory $\mathbf{x}_d(t)$ results from the summation of two wrenches acting on a mass-damper system: a linear spring wrench with stiffness k_a between the current setpoint $\mathbf{x}_d$ and a goal variable $\mathbf{g}_z$, and a predefined phase-encoded wrench $\mathbf{f}_z(\phi)$. Adjusting the parameters m_a , d_a , and k_a controls the convergence speed to the goal $\mathbf{g}_z$. Traditionally, such parameter adaptations affect predetermined phase dynamics, as seen in [21] through

$$\sqrt{m_a} \dot{\phi} = -\kappa_z \phi \quad (62)$$

where $\kappa_z \in \mathbb{R}_{\geq 0}$ is a user-defined parameter. In contrast, in the proposed guidance law (9), the applied wrench on the mass-damper system comes solely from the human-applied wrench $\mathbf{f}_h$, giving full control over speed to the human. While DMPs shape only the overall trajectory duration and keep dynamics unchanged, the proposed motion guidance law allows dynamic adjustment of progression speed along the path. Note that in (61), by setting $k_a = 0$ and replacing the predetermined wrench $\mathbf{f}_z$ with the human-applied wrench $\mathbf{f}_h$, the DMP formulation transforms into (9).

APPENDIX C

ADMITTANCE VARIATION POWER

Considering (6), the time-derivative of the phase-based admittance is

$$\dot{m}_\phi = \dot{m}_a \frac{d\mathbf{x}_d^T}{d\phi} \frac{d\mathbf{x}_d}{d\phi} + 2m_a \dot{\phi} \frac{d^2\mathbf{x}_d^T}{d\phi^2} \frac{d\mathbf{x}_d}{d\phi}. \quad (63)$$

Hence, employing (63) and (8) in (25), we have

$$\begin{aligned} \dot{m}_\phi \dot{\phi}^2 - 2c_\phi \dot{\phi}^3 &= \dot{m}_a \frac{d\mathbf{x}_d^T}{d\phi} \frac{d\mathbf{x}_d}{d\phi} \dot{\phi}^2 + 2m_a \dot{\phi}^3 \frac{d^2\mathbf{x}_d^T}{d\phi^2} \frac{d\mathbf{x}_d}{d\phi} \\ &\quad - 2m_a \frac{d^2\mathbf{x}_d^T}{d\phi^2} \frac{d\mathbf{x}_d}{d\phi} \dot{\phi}^3 \\ &= \dot{m}_a \dot{\mathbf{x}}_d^T \dot{\mathbf{x}}_d \end{aligned} \quad (64)$$

where the chain rule (24) was considered.

APPENDIX D

ADMITTANCE VARIATION RATE LIMITER

Considering $m_a(t)$ as the admittance mass determined by (17) at time t , and $m_a^*(t - \Delta t)$ as its tank-modified value in the

previous time-step, $\dot{m}_a(t)$ can be approximated by

$$\dot{m}_a(t) = \frac{m_a(t) - m_a^*(t - \Delta t)}{\Delta t}. \quad (65)$$

The admittance variation rate can then be limited by μ if $m_a^*(t)$ is modified as follows:

$$m_a^*(t) = \begin{cases} m_a(t) & \text{if } |\dot{m}_a(t)| \leq \mu \\ m_a^*(t - \Delta t) + \mu \frac{m_a(t) - m_a^*(t - \Delta t)}{|\dot{m}_a(t)|} & \text{else.} \end{cases} \quad (66)$$

The modified admittance mass variation rate is

$$\dot{m}_a^*(t) = \frac{m_a^*(t) - m_a^*(t - \Delta t)}{\Delta t} \quad (67)$$

which according to (66) becomes

$$\dot{m}_a^*(t) = \begin{cases} \dot{m}_a(t) & \text{if } |\dot{m}_a(t)| \leq \mu \\ \underbrace{\mu \frac{m_a(t) - m_a^*(t - \Delta t)}{\Delta t |\dot{m}_a(t)|}}_{\pm 1} & \text{else.} \end{cases} \quad (68)$$

As a result, the modified admittance variation rate is always limited by μ as follows:

$$|\dot{m}_a^*| \leq \mu. \quad (69)$$

ACKNOWLEDGMENT

The authors express their gratitude to H. Lehmann for assisting with the questionnaire for Experiment 5.

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